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October 13, 2025
(Amendment to IEEE Std 802.1AS™-2025)

Draft Standard for Local and metropolitan area networks—

Timing and Synchronization for Time-Sensitive Applications

Amendment: Support for the IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

Sponsor

LAN/MAN Standards Committee
of the
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Time-Sensitive Networking (TSN) Task Group of IEEE 802.1

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The text proper of this draft begins with the title page (1). The cover pages (a), (b), (c) etc. are for 802.1 WG information, and will be removed prior to Sponsor Ballot.

Editors’ Foreword

This draft standard is an amendment. The scope of changes to the base standard is thus strictly limited, as detailed in the PAR.

Information on participation in this project, and in the IEEE 802.1 Working Group can be found here.

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Project Authorization Request, Scope, Purpose, and Criteria for Standards Development (CSD)

The complete amendment PAR, as approved by IEEE NesCom 23 February 2022, can be found at:

https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/9522

The ‘Scope of the Proposed changes’ and the ‘Need for the Project’ specify the changes to be made by this amendment (see below).

Scope of the Proposed changes:

This amendment specifies protocols, procedures, and managed objects that support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex while retaining existing functionality and backward compatibility, and remaining a profile of IEEE Std 1588™-2019.
This amendment addresses errors and omissions in the description of existing functionality.

Need for the Project:

Support is needed in applications such as automotive in-vehicle networks and industrial automation networks for the IEEE Std 802.3 Clause 4 MAC operating in half-duplex, including those using links with the 10BASE-T1S PHY in either point-to-point or multidrop half-duplex mode recently introduced by IEEE Std 802.3cg-2019.

Criteria for Standards Development:

The complete Criteria for Standards Development (CSD) can be found at:

https://mentor.ieee.org/802-ec/dcn/21/ec-21-0308-00-ACSD-p802-1asds.pdf

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Prepared by the
Time-Sensitive Networking (TSN)Task Group of IEEE 802.1

Sponsor
LAN/MAN Standards Committee
of the
IEEE Computer Society

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Abstract: This amendment to IEEE Std 802.1AS™-2020 specifies protocols, procedures, and managed objects that support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex while retaining existing functionality and backward compatibility, and remaining a profile of IEEE Std 1588™-2019.

This amendment addresses errors and omissions in the description of existing functionality.

Keywords: best timeTransmitter, frequency offset, Grandmaster Clock, Grandmaster PTP Instance, PTP End Instance, PTP Relay Instance, IEEE Std 802.1AS™, phase offset, synchronization, syntonization, time-aware system

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Introduction

This introduction is not part of IEEE Std 802.1ASds™-20xx, IEEE Standard for Local and metropolitan area networks—Timing and Synchronization for Time-Sensitive Applications—Amendment: Support for the IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

The first edition of IEEE Std 802.1AS was published in 2011. A first corrigendum, IEEE Std 802.1AS™-2011/Cor1-2013, provided technical and editorial corrections. A second corrigendum, IEEE Std 802.1AS™-2011/Cor2-2015 provided additional technical and editorial corrections.

The second edition, IEEE Std 802.1AS-2020, added support for multiple gPTP domains, Common Mean Link Delay Service, external port configuration, and Fine Timing Measurement for 802.11 transport. Backward compatibility with IEEE Std 802.1AS-2011 was maintained. A corrigendum, IEEE Std 802.1AS™-2020/Cor1-2021, provides technical and editorial corrections.

The third edition, IEEE Std 802.1AS-202x is a roll-up of IEEE Std 802.1AS-2020 with the corrigendum IEEE Std 802.1AS-2020/Cor1, and its amendments: IEEE Std 802.1ASdr, IEEE Std 802.1ASdn, and IEEE Std 802.1ASdm.

This amendment to IEEE Std 802.1AS-202x specifies protocols, procedures, and managed objects that support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex while retaining existing functionality and backward compatibility, and remaining a profile of IEEE Std 1588™-2019.

This amendment addresses errors and omissions in the description of existing functionality

<<Editor’s note: P802.1ASds is an amendment to 802.1AS-2025 that is the outcome of P802.1AS-2020-REV.>>

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[This amendment is based on IEEE Std 802.1AS™-20xx (IEEE Std 802.1AS™-2020 Revision).

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strikethrough~~ (to remove old material) and <u>underscore</u> (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.[1]

[1]Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

3. Definitions

Insert the following definitions in Clause 3, and renumber the definitions as appropriate:

3.17 Half-duplex Ethernet: Half-duplex Ethernet: IEEE Std 802.3™ Clause 4 Media Access Control (MAC) operating in half-duplex.

4. Acronyms and abbreviations

Insert the following acronym in clause 4 as follows:

HDE Half-duplex Ethernet

5. Conformance

5.5 .MAC-specific timing and synchronization methods for full-duplex IEEE 802.3 links

Change the first and sentence of 5.5 as follows:

An implementation of a time-aware system with full-duplex IEEE Std 802.3-2022 media access control (MAC) services to physical ports shall:

Change the sentence after item e) of 5.5 as follows:

An implementation of a PTP Instance with full-duplex IEEE Std 802.3-2022 MAC services to physical ports may:

Insert 5.9 as follows:

5.9 MAC-specific timing and synchronization methods for half-duplex Ethernet (HDE) links

An implementation of a time-aware system with half-duplex IEEE Std 802.3-2022 media access control (MAC) services to physical ports shall:

a) Support half-duplex operation, as specified in Clause 4 of IEEE Std 802.3-2022.
b) Support the requirements as specified in Clause 19.

7. Time-synchronization model for a packet network

7.2 Architecture of a time-aware network

7.2.1 General

Add item g) after item f) in the lettered list in 7.2.1 as follows:

g) IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex as specified in Clause 19.

7.6 Differences between gPTP (IEEE Std 802.1AS) and PTP (IEEE Std 1588-2019)

Change item e) and item f) in 7.5 as follows:

e) For full-duplex Ethernet links and HDE links, gPTP requires the use of the peer-to-peer delay mechanism, while IEEE Std 1588-2019 also allows the use of end-to-end delay measurement.

f) For full-duplex Ethernet links and HDE links, gPTP requires the use of two-step processing (use of Follow_Up and Pdelay_Resp_Follow_Up messages to communicate timestamps), with an optional one-step processing mode for full-duplex Ethernet that embeds timestamps in the Sync "on the fly" as they are being transmitted (gPTP does not specify one-step processing for peer delay messages). IEEE Std 1588-2019 allows either two-step or one-step processing to be required (for both Sync and peer delay messages) depending on a specific profile.

8. IEEE 802.1AS concepts and terminology

8.5 Ports

8.5.1 General

Change 8.5.1 as follows:

The PTP Instances in a gPTP domain interface with the network media via physical ports. gPTP defines ~~a logical port, i.e.,~~ a PTP Port, in such a way that communication between PTP Instances is point-to-point or, point-to-multipoint. The choice of point-to-point (e.g., see Clause 16) or point-to-multipoint (e.g., see Clause 19) communication is specified in each media dependent specification in this standard. ~~even over physical ports that are attached to shared media. One logical~~A PTP port~~,~~ consists~~ing~~ of one PortSync entity and one media-dependent (MD) entity.~~, is instantiated for each PTP Instance with which the PTP Instance communicates. For shared media, multiple logical ports can be associated with a single physical port.~~

~~Unless otherwise qualified, each instance of the term *port* refers to a *logical port*.~~

10. Media-independent layer specification

10.2 Time-synchronization state machines

10.2.4 Per PTP Instance global variables

10.2.4.28 driftTrackingTlvSupport

Change the note in 10.2.4.28 as follows:

NOTE—The Drift_Tracking TLV is transported only on full-duplex, point-to-point links (as specified in Clause 11), and HDE links (as specified in Clause 19). Even if driftTrackingTlvSupport is TRUE, the Drift_Tracking TLV is not transported on links other than full-duplex, point-to-point links, and HDE links, and is not received by the PTP Ports at the other end of these links.

10.2.5 Per-port global variables

10.2.5.1 asCapable

Change the note in 10.2.5.1 as follows:

NOTE—For full-duplex point-to-point links, ~~The per-port global variable~~ asCapableAcrossDomains (see 11.2.13.13) is common across, and accessible by, all the domains. For HDE links, asCapableAcrossDomains is a per-PTP Instance (i.e., per domain). ~~It~~asCapableAcrossDomains is computed by the MDPdelayReq state machine (see 11.2.19). For full-duplex point-to-point links (see Clause 11) and HDE links (see Clause 19), asCapableAcrossDomains is used when setting the instance of asCapable for each domain (for the link in question).

10.5 Message attributes

10.5.3 Addresses

Change the last paragraph in 10.5.3 as follows:

If the transport is full-duplex and half-duplex IEEE 802.3, all Announce and Signaling messages transmitted shall use the MAC address of the respective egress physical port as the source address.

10.6 Message formats

10.6.4 Signaling message

10.6.4.3 Message interval request TLV definition

10.6.4.3.9 flags (Octet)

Change the note in 10.6.4.3.9 as follows:

NOTE—For full-duplex point-to-point links (see Clause 11) and HDE links (see Clause 19), it is expected that the PTP Port sending this TLV will set bits 0 and/or 1 to FALSE if this PTP Port will not provide valid timing information in its subsequent responses (Pdelay_Resp and Pdelay_Resp_Follow_Up) to Pdelay_Req messages. Similarly, it is expected that the PTP Port sending this TLV will set bit 2 to TRUE if it is capable of receiving and correctly processing one-step Sync messages.

11. Media-dependent layer specification for full-duplex point-to-point links

11.1 Overview

11.1.1 General

Add a NOTE at the end of 11.1.1 as follows:

NOTE—PTP links using the IEEE Std 802.3 Clause 4 MAC operating in half-duplex mode are specified in Clause 19.

11.2 State machines for MD entity specific to full-duplex point-to-point links

11.2.1 General

Replace Figure 11-5 with the following:

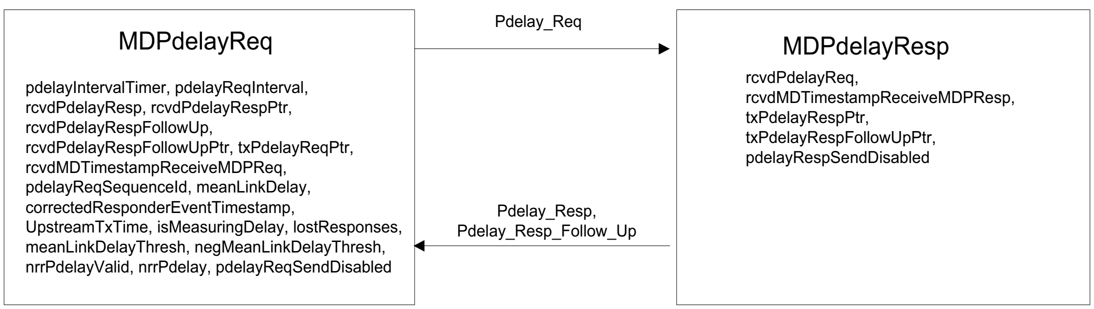


Figure 11-5—Peer-to-peer delay mechanism state machines—overview and interrelationships

11.2.2 Determination of asCapable and asCapableAcrossDomains

Change 11.2.2 as follows:

There is one instance of the global variable asCapable (see 10.2.5.1) per PTP Port, per domain. For full-duplex point-to-point links, ~~T~~there is one instance of the global variable asCapableAcrossDomains (see 11.2.13.13), per port, that is common across, and accessible by, all the domains. For HDE links, there is one instance of asCapableAcrossDomains per PTP Instance (i.e., per domain) (see 11.2.13.13).

The ~~per-PTP Port~~ global variable asCapable (see 10.2.5.1) indicates whether the IEEE 802.1AS protocol is operating, in this domain, on the PTP Link attached to this PTP Port, and can provide the required time-synchronization performance ~~described in B.3~~. asCapable is used by the PortSync entity, which is media-independent; however, the determination of asCapable is media-dependent.

The per-port global variable asCapableAcrossDomains is set by the MDPdelayReq state machine (see 11.2.19 and Figure 11-9). For a port attached to a full-duplex point-to-point PTP Link or to an HDE link, asCapableAcrossDomains shall be set to TRUE if and only if either:

a) I~~i~~t is determined, via the transport-specific peer-to-peer delay mechanism or CMLDS, that the following conditions hold for the port:
   1) ~~a)~~The port is exchanging peer delay messages with its neighbor,
   2) ~~b)~~The measured delay does not exceed meanLinkDelayThresh or is below negMeanLinkDelayThresh,
   3) ~~c)~~For full-duplex links, ~~T~~the port does not receive multiple Pdelay_Resp or Pdelay_Resp_Follow_Up messages in response to a single Pdelay_Req message, ~~and~~
   4) For HDE links, the port does not receive multiple Pdelay_Resp or multiple Pdelay_Resp_Follow_Up messages with the requestingPortIdenity.Clock Identity equal to the PTP Instance's clock identity in response to a single Pdelay_Req message, and
   5) ~~d)~~The port does not receive a response from itself or another PTP Port of the same PTP Instance.

or:

b) pdelayReqSendDisabled is set to TRUE.

NOTE 1—If a PTP Instance implements only domain 0 and the MDPdelayReq and MDPdelayResp state machines are invoked on domain 0 (see 11.2.19), asCapableAcrossDomains is still set by the MDPdelayReq state machine.

The default value of meanLinkDelayThresh shall be set as specified in Table 11-1.

Table 11-1—Value of meanLinkDelayThresh for various links

Link	Value of meanLinkDelayThresh (ns) (see NOTE)
100BASE-TX, 1000BASE-T	800_{10}
100BASE-FX, 1000BASE-X	0x7FFFFFFF
NOTE—The actual propagation delay for 100BASE-TX and 1000BASE-T links is expected to be smaller than the above respective threshold. If the measured mean propagation delay (i.e., meanLinkDelay; see 10.2.5.8) exceeds this threshold, it is assumed that this is due to the presence of equipment that does not implement gPTP. For 100BASE-FX and 1000BASE-Xlinks, the actual propagation delay can be on the order of, or larger than, the delay produced by equipment that does not implement gPTP; therefore, such equipment cannot be detected by comparing measured propagation delay with a threshold. In this case, meanLinkDelayThresh is set to the largest possible positive value (i.e., in binary notation a zero followed by all 1s). The default value for negMeanLinkDelayThresh is not defined and it is implementation specific. This table does not list propagation delay threshold values for all types of PHYs because the values are implementation specific and depend on the application.	

The per-PTP Port, per-domain global variable asCapable shall be set to TRUE if and only if the following conditions hold:

c) ~~e)~~The value of asCapableAcrossDomains is TRUE, and
d) ~~f)~~One of the following conditions holds:
   1) The value of neighborGptpCapable for this PTP Port is TRUE, or
   2) The value of domainNumber is zero, and the value of sdoId for peer delay messages received on this PTP Port is 0x100.

NOTE 2—Condition ~~f)~~ d) 2) ensures backward compatibility with the 2011 edition of this standard. A PTP Instance compliant with the current edition of this standard that is attached, via a full-duplex point-to-point PTP Link, to a PTP

Instance compliant with the 2011 edition of this standard will not receive Signaling messages that contain the gPTP capable TLV and will not set neighborGptpCapable to TRUE. However, condition ~~f)~~ d) 2) ensures that asCapable for this PTP Port and domain (i.e., domain 0) will still be set to TRUE if condition c) holds because the peer delay messages received from the time-aware system compliant with the 2011 edition of this standard will have sdoId set to 0x100.

11.2.12 Overview of MD entity global variables

Add the following note in Table 11-2:

NOTE — asCapableAcrossDomains is per PTP Instance for HDE links.

11.2.13 MD entity global variables

Change 11.2.13.13 as follows:

11.2.13.13 asCapableAcrossDomains: A Boolean that is TRUE if and only if either: 11.2.2 item a) or 11.2.2 item b) is satisfied~~conditions a) through d) of 11.2.2 are satisfied~~. ~~This Boolean~~ asCapableAcrossDomains is set by the MDPdelayReq state machine and is used in determining asCapable for a port (see 11.2.2). There is one instance of this variable for all the domains (per port) for full-duplex point-to-point links,~~. T~~and the variable is accessible by all the domains. There is one instance of this variable per PTP Instance (i.e., per domain) for HDE links (see 19.2.2). When only one domain is active, asCapableAcrossDomains is equivalent to the variable asCapable (see 10.2.5.1).

11.2.19 MDPdelayReq state machine

11.2.19.2 State machine variables

Change 11.2.19.2.2 as follows:

11.2.19.2.2 rcvdPdelayResp:

a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp message is received. This variable is reset by the current state machine.

b) For half-duplex mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp message is received AND its requestingPortIdentity.clockIdentity is equal to the clockIdentity of the current PTP Instance. This variable is reset by the current state machine.

Change 11.2.19.2.4 as follows:

11.2.19.2.4 rcvdPdelayRespFollowUp:

a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp_Follow_Up message is received. This variable is reset by the current state machine.

b) For half-duplex mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp_Follow_Up message is received AND its requestingPortIdentity.clockIdentity is equal to the clockIdentity of the current PTP instance. This variable is reset by the current state machine.

Insert a new variable after 11.2.19.2.13 as follows:

11.2.19.2.14 pdelayReqSendDisabled: A boolean that is administratively set to TRUE if Pdelay_Req messages are not transmitted by this port. The default value for this variable shall be FALSE.

11.2.19.4 State diagram

Replace Figure 11-9 with the following:.

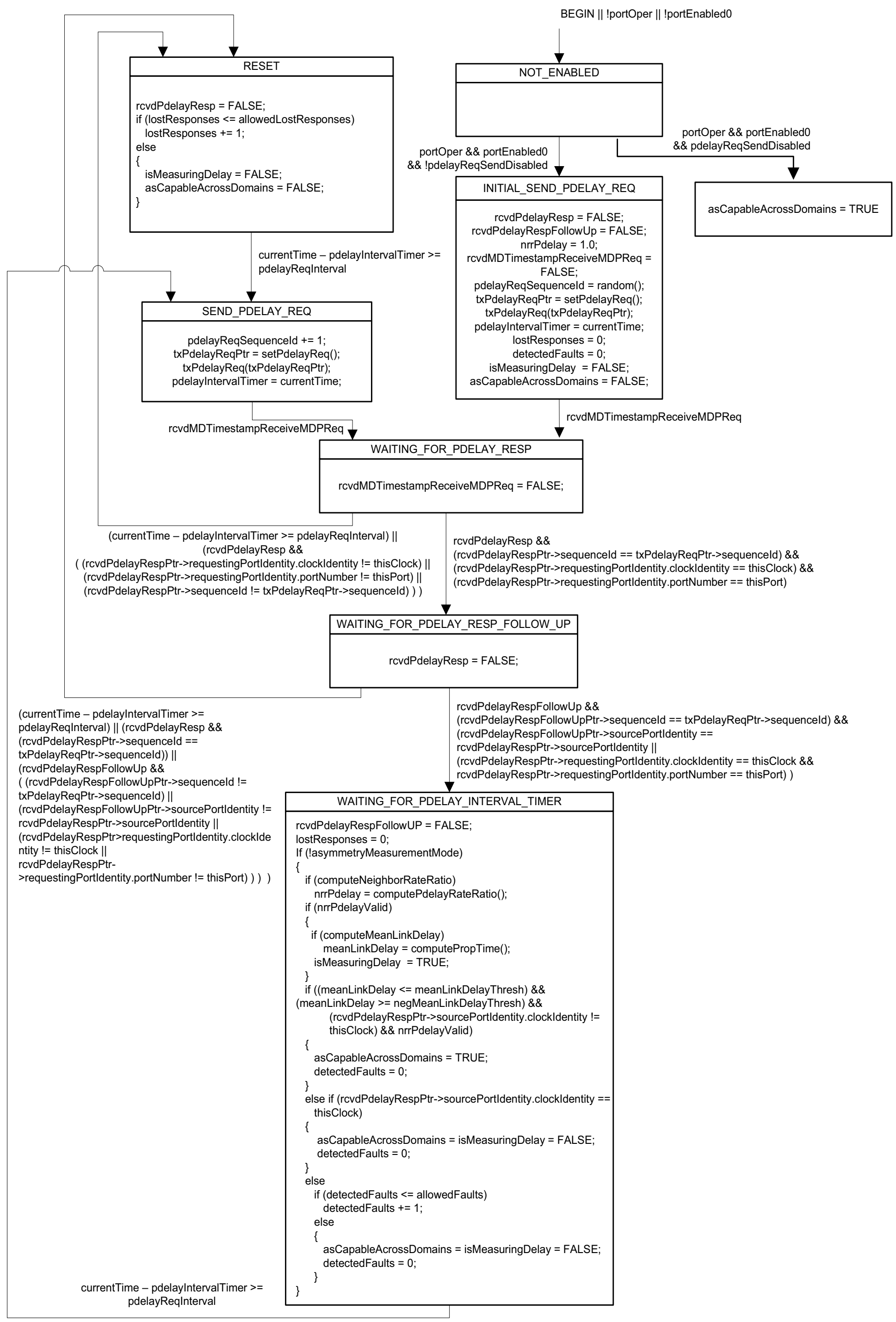


Figure 11-9—MDPdelayReq state machine

Add a NOTE after Figure 11-9 as follows:

NOTE—A change in the value of the variable pdelayReqSendDisabled takes effect only when portEnabled0 (see 11.2.19.2.12) is FALSE.

11.2.20 MDPdelayResp state machine

11.2.20.2 State machine variables

Insert a new variable after 11.2.20.2.5 as follows:

11.2.20.2.6 pdelayRespSendDisabled: A boolean that is administratively set to TRUE to indicate that corresponding Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by this port in response to a Pdelay_Req message. The default value for this variable shall be FALSE.

11.2.20.4 State diagram

.Replace Figure 11-10 with the following:

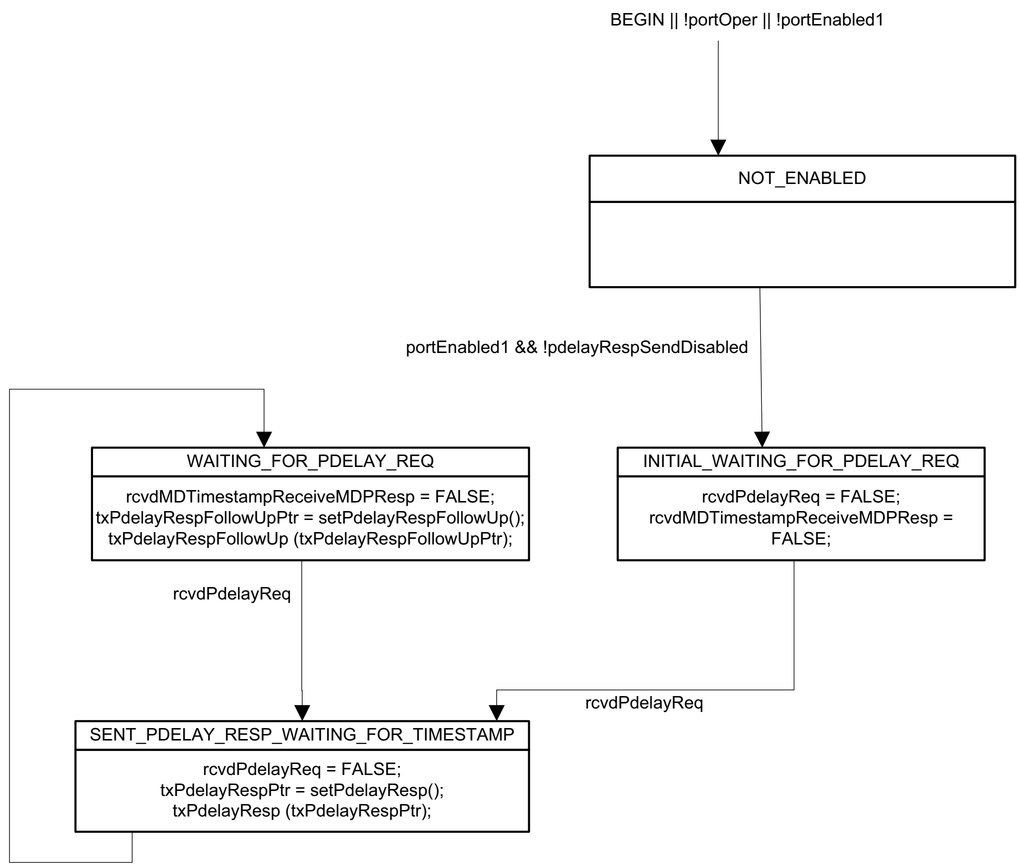


Figure 11-10—MDPdelayResp state machine

Add a NOTE after Figure 11-10 as follows:

NOTE—A change in the value of the variable pdelayRespSendDisabled takes effect only when portEnabled1 (see 11.2.20.2.5) is FALSE.

11.4 Message formats

11.4.2 Header

11.4.2.4 domainNumber (UInteger8)

Change the paragraph in 11.4.2.4 as follows:

For full-duplex, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmitted~~The~~ with domainNumber ~~for Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be~~ set to 0. For HDE, see 19.2.19 and 19.2.20. The domainNumber for all other PTP messages is as specified in 10.6.2.2.6.

14. Timing and synchronization management

14.10 Port Parameter Data Set (portDS)

14.10.1 General

Insert the following items after the final item of Table 14-9:

Table 14-9—portDS table

Name	Data type	Operations supported[a]	References
pdelayReqSendDisabled	Boolean	RW	14.10.60
pdelayRespSendDisabled	Boolean	RW	14.10.61

[a] R = Read only access; RW = Read/write access.

14.10.23 InitialLogPdelayReqInterval

Change the first paragraph in 14.10.23 as follows:

For full-duplex and half-duplex IEEE Std 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to base 2 of the Pdelay_Req message transmission interval used when:

14.10.24 currentLogPdelayReqInterval

Change the first paragraph in 14.10.24 as follows:

For full-duplex and half-duplex IEEE Std 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to the base 2 of the current Pdelay_Req message transmission interval (see 11.5.2.2).

14.10.32 currentComputeNeighborRateRatio

Change the first paragraph in 14.10.32 as follows:

For full-duplex and half-duplex IEEE Std 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the current value of computeNeighborRateRatio.

14.10.36 currentComputeMeanLinkDelay

Change the first paragraph in 14.10.36 as follows:

For full-duplex and half-duplex IEEE Std 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the current value of computeMeanLinkDelay.

14.10.53 pdelayTruncatedTimestampsArray

Change the first paragraph in 14.10.53 as follows:

For full-duplex and half-duplex IEEE Std 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the values of the four elements of this array are as described in Table 14-9. For all other media, the values are zero. Array elements 0, 1, 2, and 3 correspond to the timestamps t1, t2, t3, and t4, respectively, in Figure 11-1 and are expressed in units of 2–16 ns (i.e., the value of each array element is equal to the remainder obtained upon dividing the respective timestamp, expressed in units of 2–16 ns, by 248). At any given time, the timestamp values stored in the array are for the same, and most recently completed, peer delay message exchange.

Insert 14.10.60 and 14.10.61 as follows:

14.10.60 pdelayReqSendDisabled

The value is equal to the value of the per-PTP Port variable pdelayReqSendDisabled (see 11.2.19.2.14). If its value is TRUE, Pdelay_Req messages are not transmitted by the PTP Port. The default value for this variable shall be FALSE.

14.10.61 pdelayRespSendDisabled

The value is equal to the value of the per-PTP Port variable pdelayRespSendDisabled (see 11.2.20.2.6). If its value is TRUE, Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by the PTP Port in response to Pdelay_Req message. The default value for this variable shall be FALSE.

17. YANG Data Model

17.2 IEEE 802.1AS YANG models

Replace Figure 17-3 with the following:

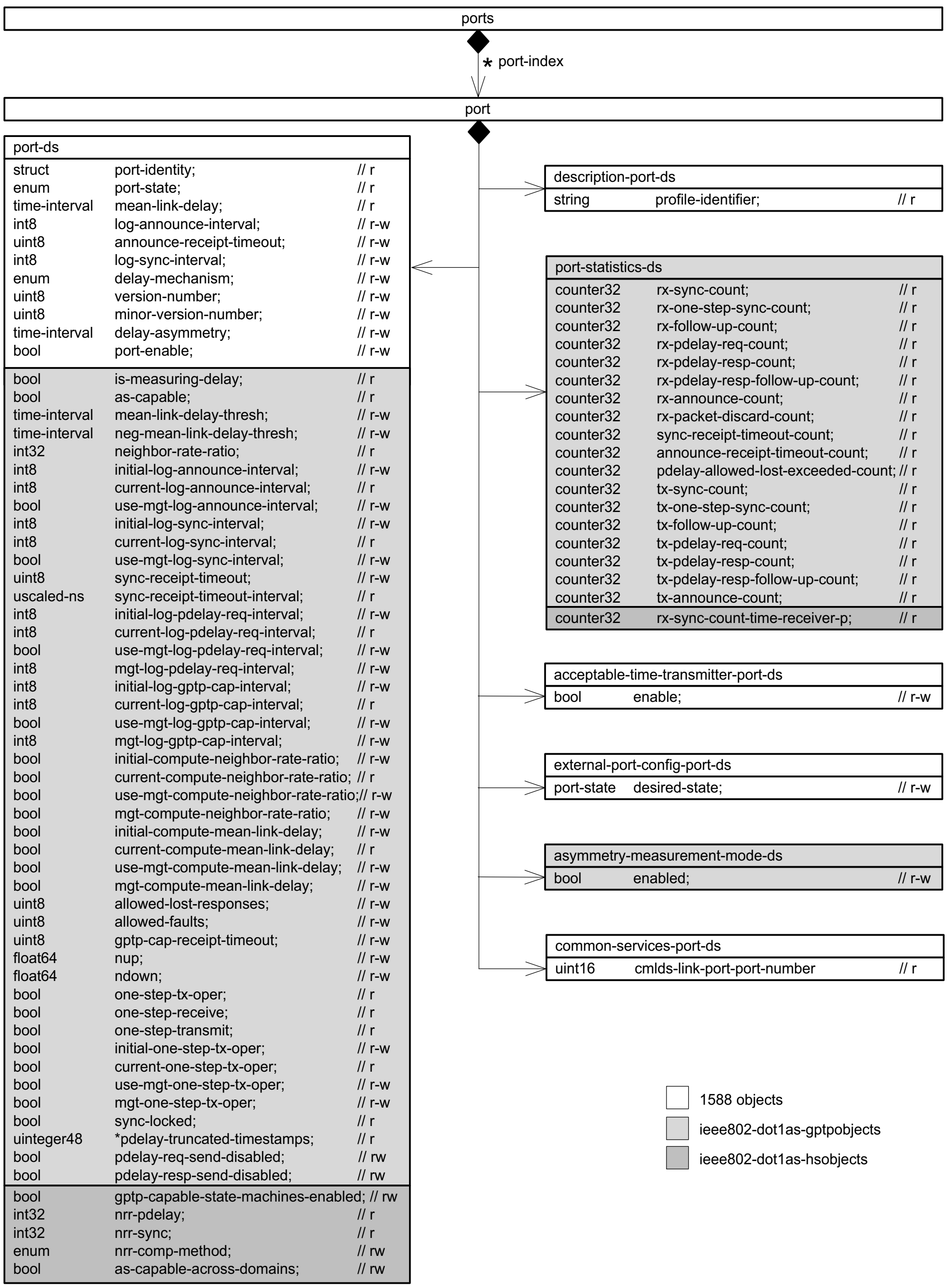


Figure 17-3—PTP Port detail

17.5 YANG schema tree definitions

17.5.1 lTree diagram for ieee802-dot1as-gptp.yang

Replace the tree diagram for ieee802-dot1as-gptp.yang as follows:

```
module: ieee802-dot1as-gptp

  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:default-ds:
    +--ro gm-capable?                 boolean
    +--ro current-utc-offset?         int16
    +--ro current-utc-offset-valid?   boolean
    +--ro leap59?                     boolean
    +--ro leap61?                     boolean
    +--ro time-traceable?             boolean
    +--ro frequency-traceable?        boolean
    +--ro ptp-timescale?              boolean
    +--ro time-source?                identityref
  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:current-ds:
    +--ro last-gm-phase-change?        scaled-ns
    +--ro last-gm-freq-change?         float64
    +--ro gm-timebase-indicator?       uint16
    +--ro gm-change-count?             yang:counter32
    +--ro time-of-last-gm-change?      yang:timestamp
    +--ro time-of-last-phase-change?   yang:timestamp
    +--ro time-of-last-freq-change?    yang:timestamp
  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds:
    +--ro cumulative-rate-ratio?   int32
  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port/
ptp-tt:port-ds:
    +--ro is-measuring-delay?                     boolean
    +--ro as-capable?                             boolean
    +--rw mean-link-delay-thresh?                 ptp-tt:time-interval
    +--rw neg-mean-link-delay-thresh?             ptp-tt:time-interval
    +--ro neighbor-rate-ratio?                    int32
    +--rw initial-log-announce-interval?          int8
    +--ro current-log-announce-interval?          int8
    +--rw use-mgt-log-announce-interval?          boolean
    +--rw initial-log-sync-interval?              int8
    +--ro current-log-sync-interval?              int8
    +--rw use-mgt-log-sync-interval?              boolean
    +--rw sync-receipt-timeout?                   uint8
    +--ro sync-receipt-timeout-interval?          uscaled-ns
    +--rw initial-log-pdelay-req-interval?        int8
    +--ro current-log-pdelay-req-interval?        int8
    +--rw use-mgt-log-pdelay-req-interval?        boolean
    +--rw mgt-log-pdelay-req-interval?            int8
    +--rw initial-log-gptp-cap-interval?          int8
    +--ro current-log-gptp-cap-interval?          int8
    +--rw use-mgt-log-gptp-cap-interval?          boolean
    +--rw mgt-log-gptp-cap-interval?              int8
    +--rw initial-compute-neighbor-rate-ratio?    boolean
    +--ro current-compute-neighbor-rate-ratio?    boolean
    +--rw use-mgt-compute-neighbor-rate-ratio?    boolean
    +--rw mgt-compute-neighbor-rate-ratio?        boolean
    +--rw initial-compute-mean-link-delay?        boolean
    +--ro current-compute-mean-link-delay?        boolean
    +--rw use-mgt-compute-mean-link-delay?        boolean
```

```
    +--rw mgt-compute-mean-link-delay?           boolean
    +--rw allowed-lost-responses?                uint8
    +--rw allowed-faults?                        uint8
    +--rw gptp-cap-receipt-timeout?              uint8
    +--rw nup?                                   float64
    +--rw ndown?                                 float64
    +--ro one-step-tx-oper?                      boolean
    +--ro one-step-receive?                      boolean
    +--ro one-step-transmit?                     boolean
    +--rw initial-one-step-tx-oper?              boolean
    +--ro current-one-step-tx-oper?              boolean
    +--rw use-mgt-one-step-tx-oper?              boolean
    +--rw mgt-one-step-tx-oper?                  boolean
    +--ro sync-locked?                           boolean
    +--ro pdelay-truncated-timestamps*           uinteger48
    +--rw pdelay-req-send-disabled?              boolean
    +--rw pdelay-resp-send-disabled?             boolean
  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
    +--rw port-statistics-ds
       +--ro rx-sync-count?                        yang:counter32
       +--ro rx-one-step-sync-count?               yang:counter32
       +--ro rx-follow-up-count?                   yang:counter32
       +--ro rx-pdelay-req-count?                  yang:counter32
       +--ro rx-pdelay-resp-count?                 yang:counter32
       +--ro rx-pdelay-resp-follow-up-count?       yang:counter32
       +--ro rx-announce-count?                    yang:counter32
       +--ro rx-packet-discard-count?              yang:counter32
       +--ro sync-receipt-timeout-count?           yang:counter32
       +--ro announce-receipt-timeout-count?       yang:counter32
       +--ro pdelay-allowed-lost-exceeded-count?   yang:counter32
       +--ro tx-sync-count?                        yang:counter32
       +--ro tx-one-step-sync-count?               yang:counter32
       +--ro tx-follow-up-count?                   yang:counter32
       +--ro tx-pdelay-req-count?                  yang:counter32
       +--ro tx-pdelay-resp-count?                 yang:counter32
       +--ro tx-pdelay-resp-follow-up-count?       yang:counter32
       +--ro tx-announce-count?                    yang:counter32
  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
    +--rw asymmetry-measurement-mode-ds
       +--rw enabled?   boolean
     augment   /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlds/ptp-tt:ports/ptp-
tt:port/ptp-tt:link-port-ds:
    +--ro cmlds-link-port-enabled?               boolean
    +--ro is-measuring-delay?                    boolean
    +--ro as-capable-across-domains?             boolean
    +--rw mean-link-delay-thresh?                ptp-tt:time-interval
    +--rw neg-mean-link-delay-thresh?            ptp-tt:time-interval
    +--rw initial-log-pdelay-req-interval?       int8
    +--ro current-log-pdelay-req-interval?       int8
    +--rw use-mgt-log-pdelay-req-interval?       boolean
    +--rw mgt-log-pdelay-req-interval?           int8
    +--rw initial-compute-neighbor-rate-ratio?   boolean
    +--ro current-compute-neighbor-rate-ratio?   boolean
    +--rw use-mgt-compute-neighbor-rate-ratio?   boolean
    +--rw mgt-compute-neighbor-rate-ratio?       boolean
    +--rw initial-compute-mean-link-delay?       boolean
    +--ro current-compute-mean-link-delay?       boolean
    +--rw use-mgt-compute-mean-link-delay?       boolean
    +--rw mgt-compute-mean-link-delay?           boolean
```

```
    +--rw allowed-lost-responses?                uint8
    +--rw allowed-faults?                        uint8
    +--ro pdelay-truncated-timestamps*           uinteger48
  augment  /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlds/ptp-tt:ports/ptp-
tt:port:
    +--rw port-statistics-ds
       +--ro rx-pdelay-req-count?                  yang:counter32
       +--ro rx-pdelay-resp-count?                 yang:counter32
       +--ro rx-pdelay-resp-follow-up-count?       yang:counter32
       +--ro rx-packet-discard-count?              yang:counter32
       +--ro pdelay-allowed-lost-exceeded-count?   yang:counter32
       +--ro tx-pdelay-req-count?                  yang:counter32
       +--ro tx-pdelay-resp-count?                 yang:counter32
       +--ro tx-pdelay-resp-follow-up-count?       yang:counter32
  augment  /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlds/ptp-tt:ports/ptp-
tt:port:
    +--rw asymmetry-measurement-mode-ds
       +--rw enabled?   boolean
```

17.6 YANG modules[1] [2]

17.6.1 Module ieee802-dot1as-gptp.yang

Replace the YANG module for ieee802-dot1as-gptp.yang as follows:

```
module ieee802-dot1as-gptp {
  yang-version 1.1;
  namespace “urn:ieee:std:802.1AS:yang:ieee802-dot1as-gptp”;
  prefix dot1as-gptp;

  import ietf-yang-types {
    prefix yang;
  }
  import ieee1588-ptp-tt {
    prefix ptp-tt;
  }

  organization
    “IEEE 802.1 Working Group”;
  contact
    “WG-URL: http://ieee802.org/1/
     WG-EMail: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
             Postal: C/O IEEE 802.1 Working Group
             IEEE Standards Association
             445 Hoes Lane
             Piscataway, NJ 08854
             USA

     E-mail: stds-802-1-chairs@ieee.org”;
  description
```

[1] Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

[2] An ASCII version of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at https://1.ieee802.org/yang-modules/.

```
    “Management objects that control timing and synchronization for time
     sensitive applications, as specified in clause 14 of IEEE Std
     802.1AS-2020-REV.

     Copyright (C) IEEE (2025). This version of this YANG module is part
       of IEEE Std 802.1AS-2020-REV; see the standard itself for full legal
notices.”;

  revision 2025-03-06 {
    description
      “Published as part of IEEE Std 802.1ASds-202X.”;
    reference
      “IEEE Std 802.1AS - Timing and Synchronization for Time-Sensitive
       Applications: IEEE Std 802.1AS-2020-REV.
       IEEE Std 1588 - IEEE Standard for a Precision Clock Synchronization
       Protocol for Networked Measurement and Control Systems: IEEE Std
       1588-2019, IEEE Std 1588g-2022, IEEE Std 1588e-2024.”;
  }

  typedef scaled-ns {
    type string {
      pattern ‘[0-9A-F]{2}(-[0-9A-F]{2}){11}’;
    }
    description
      “The IEEE Std 802.1AS ScaledNs type represents signed values of
       time and time interval in units of 2^16 ns, as a signed 96-bit
       integer. Each of the 12 octets is represented as a pair of
       hexadecimal characters, using uppercase for a letter. Octets are
       separated by a dash character. The most significant octet is first.”;
    reference
      “6.4.3.1 of IEEE Std 802.1AS”;
  }

  typedef uscaled-ns {
    type string {
      pattern ‘[0-9A-F]{2}(-[0-9A-F]{2}){11}’;
    }
    description
      “The IEEE Std 802.1AS UScaledNs type represents unsigned values of
       time and time interval in units of 2^16 ns, as an unsigned 96-bit
       integer. Each of the 12 octets is represented as a pair of
       hexadecimal characters, using uppercase for a letter. Octets are
       separated by a dash character. The most significant octet is first.”;
    reference
      “6.4.3.2 of IEEE Std 802.1AS”;
  }

  typedef float64 {
    type string {
      pattern ‘[0-9A-F]{2}(-[0-9A-F]{2}){7}’;
    }
    description
      “The IEEE Std 802.1AS Float64 type represents IEEE Std 754
       binary64. Each of the 8 octets is represented as a pair of
       hexadecimal characters, using uppercase for a letter. Octets are
       separated by a dash character. The most significant octet is first.”;
    reference
      “6.4.2 of IEEE Std 802.1AS”;
  }
```

```
typedef uinteger48 {
  type uint64 {
    range “0..281474976710655”;
  }
  description
    “48-bit unsigned integer data type.”;
  reference
    “6.4.2 of IEEE Std 802.1AS”;
}

augment “/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance”
      + “/ptp-tt:default-ds” {
  description
    “Augment IEEE Std 1588 defaultDS.”;
  leaf gm-capable {
    type boolean;
    config false;
    description
      “The value is true if the time-aware system is capable of being a
       Grandmaster, and false if the time-aware system is not capable of
       being a Grandmaster.”;
    reference
      “14.2.7 of IEEE Std 802.1AS”;
  }
  leaf current-utc-offset {
    when “../current-utc-offset-valid=’true’”;
    type int16;
    config false;
    description
      “Offset from UTC (TAI - UTC). The offset is in units of seconds.
       This leaf applies to the ClockTimeTransmitter entity (i.e., local
       only, unrelated to a remote GM).”;
    reference
      “14.2.8 of IEEE Std 802.1AS”;
  }
  leaf current-utc-offset-valid {
    type boolean;
    config false;
    description
      “The value of current-utc-offset-valid shall be true if the value
       of current-utc-offset is known to be correct, otherwise it shall
       be false. This leaf applies to the ClockTimeTransmitter entity
       (i.e., local only, unrelated to a remote GM).”;
    reference
      “14.2.9 of IEEE Std 802.1AS”;
  }
  leaf leap59 {
    type boolean;
    config false;
    description
      “If the timescale is PTP, a true value for leap59 shall indicate
       that the last minute of the current UTC day contains 59 seconds.
       If the timescale is not PTP, the value shall be false. This leaf
       applies to the ClockTimeTransmitter entity (i.e., local only,
       unrelated to a remote GM).”;
    reference
      “14.2.10 of IEEE Std 802.1AS”;
  }
```

```
leaf leap61 {
  type boolean;
  config false;
  description
    “If the timescale is PTP, a true value for leap61 shall indicate
     that the last minute of the current UTC day contains 61 seconds.
     If the timescale is not PTP, the value shall be false. This leaf
     applies to the ClockTimeTransmitter entity (i.e., local only,
     unrelated to a remote GM).”;
  reference
    “14.2.11 of IEEE Std 802.1AS”;
}
leaf time-traceable {
  type boolean;
  config false;
  description
    “The value of time-traceable shall be true if the timescale is
     traceable to a primary reference; otherwise, the value shall be
     false. This leaf applies to the ClockTimeTransmitter entity
     (i.e., local only, unrelated to a remote GM).”;
  reference
    “14.2.12 of IEEE Std 802.1AS”;
}
leaf frequency-traceable {
  type boolean;
  config false;
  description
    “The value of frequency-traceable shall be true if the frequency
     determining the timescale is traceable to a primary reference;
     otherwise, the value shall be false. This leaf applies to the
     ClockTimeTransmitter entity (i.e., local only, unrelated to a
     remote GM).”;
  reference
    “14.2.13 of IEEE Std 802.1AS”;
}
leaf ptp-timescale {
  type boolean;
  config false;
  description
    “If ptp-timescale is true, the timescale of the
     ClockTimeTransmitter entity is PTP, which is the elapsed time
     since the PTP epoch measured using the second defined by
     International Atomic Time (TAI). If ptp-timescale is false, the
     timescale of the ClockTimeTransmitter entity is ARB, which is the
     elapsed time since an arbitrary epoch. This leaf applies to the
     ClockTimeTransmitter entity (i.e., local only, unrelated to a
     remote GM).”;
  reference
    “14.2.14 of IEEE Std 802.1AS”;
}
leaf time-source {
  type identityref {
    base ptp-tt:time-source;
  }
  config false;
  description
    “The source of time used by the Grandmaster Clock. This leaf
     applies to the ClockTimeTransmitter entity (i.e., local only,
     unrelated to a remote GM).”;
```

```
    reference
      “14.2.15 of IEEE Std 802.1AS”;
  }
}

augment “/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance”
      + “/ptp-tt:current-ds” {
  description
    “Augment IEEE Std 1588 currentDS.”;
  leaf last-gm-phase-change {
    type scaled-ns;
    config false;
    description
      “Phase change that occurred on the most recent change in either
       the Grandmaster PTP Instance or gm-timebase-indicator leaf.”;
    reference
      “14.3.4 of IEEE Std 802.1AS”;
  }
  leaf last-gm-freq-change {
    type float64;
    config false;
    description
      “Frequency change that occurred on the most recent change in
       either the Grandmaster PTP Instance or gm-timebase-indicator
       leaf.”;
    reference
      “14.3.5 of IEEE Std 802.1AS”;
  }
  leaf gm-timebase-indicator {
    type uint16;
    config false;
    description
      “The timeBaseIndicator of the current Grandmaster PTP Instance.”;
    reference
      “14.3.6 of IEEE Std 802.1AS”;
  }
  leaf gm-change-count {
    type yang:counter32;
    config false;
    description
      “This statistics counter tracks the number of times the
       Grandmaster PTP Instance has changed in a gPTP domain.”;
    reference
      “14.3.7 of IEEE Std 802.1AS”;
  }
  leaf time-of-last-gm-change {
    type yang:timestamp;
    config false;
    description
      “System time when the most recent Grandmaster Clock change
       occurred in a gPTP domain. This leaf’s type is YANG timestamp,
       which is based on system time. System time is an unsigned integer
       in units of 10 milliseconds, using an epoch defined by the
       implementation (typically time of boot-up).”;
    reference
      “14.3.8 of IEEE Std 802.1AS”;
  }
  leaf time-of-last-phase-change {
    type yang:timestamp;
```

```
    config false;
    description
      “System time when the most recent change in Grandmaster Clock
       phase occurred. This leaf’s type is YANG timestamp, which is
       based on system time. System time is an unsigned integer in units
       of 10 milliseconds, using an epoch defined by the implementation
       (typically time of boot-up).”;
    reference
      “14.3.9 of IEEE Std 802.1AS”;
  }
  leaf time-of-last-freq-change {
    type yang:timestamp;
    config false;
    description
      “System time when the most recent change in Grandmaster Clock
       frequency occurred. This leaf’s type is YANG timestamp, which is
       based on system time. System time is an unsigned integer in units
       of 10 milliseconds, using an epoch defined by the implementation
       (typically time of boot-up).”;
    reference
      “14.3.10 of IEEE Std 802.1AS”;
  }
}

augment “/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance”
      + “/ptp-tt:parent-ds” {
  description
    “Augment IEEE Std 1588 parentDS.”;
  leaf cumulative-rate-ratio {
    type int32;
    config false;
    description
      “Estimate of the ratio of the frequency of the Grandmaster Clock
       to the frequency of the LocalClock entity of this PTP Instance.
       cumulative-rate-ratio is expressed as the fractional frequency
       offset multiplied by 2^41, i.e., the quantity (rateRatio -
       1.0)(2^41).”;
    reference
      “14.4.3 of IEEE Std 802.1AS”;
  }
}

augment “/ptp-tt:ptp”
      + “/ptp-tt:instances”
      + “/ptp-tt:instance”
      + “/ptp-tt:ports”
      + “/ptp-tt:port”
      + “/ptp-tt:port-ds” {
  description
    “Augment IEEE Std 1588 portDS.

     14.10.4 of IEEE Std 802.1AS specifies ptpPortEnabled
     (ptp-port-enabled), which is provided in YANG as the semantically
     equivalent node in ieee1588-ptp-tt named port-enable (in port-ds).

     14.10.15 of IEEE Std 802.1AS specifies
     mgtSettableLogAnnounceInterval (mgt-log-announce-interval), which
     is provided in YANG as the semantically equivalent node in
     ieee1588-ptp-tt named log-announce-interval (in port-ds). In the
```

```
      context of IEEE Std 802.1AS, log-announce-interval cannot be used
      unless use-mgt-log-announce-interval is true.

      14.10.20 of IEEE Std 802.1AS specifies mgtSettableLogSyncInterval
      (mgt-log-sync-interval), which is provided in YANG as the
      semantically equivalent node in ieee1588-ptp-tt named
      log-sync-interval (in port-ds). In the context of IEEE Std 802.1AS,
      log-sync-interval cannot be used unless use-mgt-log-sync-interval
      is true.”;
   leaf is-measuring-delay {
     type boolean;
     config false;
     description
       “Boolean that is true if the port is measuring PTP Link
        propagation delay.”;
     reference
       “14.10.6 of IEEE Std 802.1AS”;
   }
   leaf as-capable {
     type boolean;
     config false;
     description
       “Boolean that is true if and only if it is determined that this
        PTP Instance and the PTP Instance at the other end of the link
        attached to this port can interoperate with each other via the
        IEEE Std 802.1AS protocol.”;
     reference
       “10.2.5.1 of IEEE Std 802.1AS
        14.10.7 of IEEE Std 802.1AS”;
   }
   leaf mean-link-delay-thresh {
     type ptp-tt:time-interval;
     description
       “Propagation time threshold for mean-link-delay, above which a
        port is not considered capable of participating in the IEEE Std
        802.1AS protocol.”;
     reference
       “14.10.9 of IEEE Std 802.1AS”;
   }
   leaf neg-mean-link-delay-thresh {
     type ptp-tt:time-interval;
     description
       “The negative propagation time threshold for mean-link-delay,
        below which a port is not considered capable of participating
        in the IEEE Std 802.1AS protocol.”;
     reference
       “14.10.10 of IEEE Std 802.1AS”;
   }
   leaf neighbor-rate-ratio {
     type int32;
     config false;
     description
       “Estimate of the ratio of the frequency of the LocalClock entity
        of the PTP Instance at the other end of the link attached to this
        PTP Port, to the frequency of the LocalClock entity of this PTP
        Instance. neighbor-rate-ratio is expressed as the fractional
        frequency offset multiplied by 2^41, i.e., the quantity
        (rateRatio - 1.0)(2^41).”;
     reference
```

```
        "14.10.12 of IEEE Std 802.1AS";
    }
    leaf initial-log-announce-interval {
      type int8;
      description
        "When use-mgt-log-announce-interval is false (i.e., change with
         Signaling message), this is the logarithm to base 2 of the
         announce interval used when the port is initialized.";
      reference
        "14.10.13 of IEEE Std 802.1AS";
    }
    leaf current-log-announce-interval {
      type int8;
      config false;
      description
        "Logarithm to base 2 of the current announce interval.";
      reference
        "14.10.14 of IEEE Std 802.1AS";
    }
    leaf use-mgt-log-announce-interval {
      type boolean;
      description
        "Boolean that determines the source of the announce interval. If
         the value is true, the announce interval
         (current-log-announce-interval) is set equal to the value of
         mgt-log-announce-interval. If the value is false, the announce
         interval is determined by the AnnounceIntervalSetting state
         machine (i.e., changed with Signaling message).";
      reference
        "14.10.15 of IEEE Std 802.1AS";
    }
    leaf initial-log-sync-interval {
      type int8;
      description
        "When use-mgt-log-sync-interval is false (i.e., change with
         Signaling message), this is the logarithm to base 2 of the
         sync interval used when the port is initialized.";
      reference
        "14.10.17 of IEEE Std 802.1AS";
    }
    leaf current-log-sync-interval {
      type int8;
      config false;
      description
        "Logarithm to base 2 of the current sync interval.";
      reference
        "14.10.18 of IEEE Std 802.1AS";
    }
    leaf use-mgt-log-sync-interval {
      type boolean;
      description
        "Boolean that determines the source of the sync interval. If the
         value is true, the sync interval (current-log-sync-interval) is
         set equal to the value of mgt-log-sync-interval. If the value is
         false, the sync interval is determined by the SyncIntervalSetting
         state machine (i.e., changed with Signaling message).";
      reference
        "14.10.19 of IEEE Std 802.1AS";
    }
```

```
leaf sync-receipt-timeout {
  type uint8;
  description
    “Number of sync intervals that a timeReceiver port waits without
     receiving synchronization information, before assuming that the
     timeTransmitter is no longer transmitting synchronization
     information and that the BTCA needs to be run, if appropriate.”;
  reference
    “14.10.21 of IEEE Std 802.1AS”;
}
leaf sync-receipt-timeout-interval {
  type uscaled-ns;
  config false;
  description
    “Time interval after which sync receipt timeout occurs if
     time-synchronization information has not been received during the
     interval.”;
  reference
    “14.10.22 of IEEE Std 802.1AS”;
}
leaf initial-log-pdelay-req-interval {
  type int8;
  description
    “When use-mgt-log-pdelay-req-interval is false (i.e., change with
     Signaling message), this is the logarithm to base 2 of the
     Pdelay_Req transmit interval used when the port is initialized.”;
  reference
    “14.10.23 of IEEE Std 802.1AS”;
}
leaf current-log-pdelay-req-interval {
  type int8;
  config false;
  description
    “Logarithm to base 2 of the current Pdelay_Req transmit interval.”;
  reference
    “14.10.24 of IEEE Std 802.1AS”;
}
leaf use-mgt-log-pdelay-req-interval {
  type boolean;
  description
    “Boolean that determines the source of the Pdelay_Req transmit
     interval. If the value is true, the Pdelay_Req transmit interval
     (current-log-pdelay-req-interval) is set equal to the value of
     mgt-log-pdelay-req-interval. If the value is false, the
     Pdelay_Req transmit interval is determined by the
     LinkDelayIntervalSetting state machine (i.e., changed with
     Signaling message).”;
  reference
    “14.10.25 of IEEE Std 802.1AS”;
}
leaf mgt-log-pdelay-req-interval {
  type int8;
  description
    “Logarithm to base 2 of the Pdelay_Req transmit interval, used if
     use-mgt-log-pdelay-req-interval is true. This value is not used
     if use-mgt-log-pdelay-req-interval is false.”;
  reference
    “14.10.26 of IEEE Std 802.1AS”;
}
```

```
leaf initial-log-gptp-cap-interval {
  type int8;
  description
    “When use-mgt-log-gptp-cap-interval is false (i.e., change with
     Signaling message), this is the logarithm to base 2 of the
     gPTP capable message interval used when the port is initialized.”;
  reference
    “14.10.27 of IEEE Std 802.1AS”;
}
leaf current-log-gptp-cap-interval {
  type int8;
  config false;
  description
    “Logarithm to base 2 of the current gPTP capable message
     interval.”;
  reference
    “14.10.28 of IEEE Std 802.1AS”;
}
leaf use-mgt-log-gptp-cap-interval {
  type boolean;
  description
    “Boolean that determines the source of the gPTP capable message
     interval. If the value is true, the gPTP capable message interval
     (current-log-gptp-cap-interval) is set equal to the value of
     mgt-gptp-cap-req-interval. If the value is false, the gPTP
     capable message interval is determined by the
     GptpCapableMessageIntervalSetting state machine (i.e., changed
     with Signaling message).”;
  reference
    “14.10.29 of IEEE Std 802.1AS”;
}
leaf mgt-log-gptp-cap-interval {
  type int8;
  description
    “Logarithm to base 2 of the gPTP capable message interval, used
     if use-mgt-log-gptp-cap-interval is true. This value is not used
     if use-mgt-log-pdelay-req-interval is false.”;
  reference
    “14.10.30 of IEEE Std 802.1AS”;
}
leaf initial-compute-neighbor-rate-ratio {
  type boolean;
  description
    “When use-mgt-compute-neighbor-rate-ratio is false (i.e., change
     with Signaling message), this is the initial value of
     computeNeighborRateRatio.”;
  reference
    “14.10.31 of IEEE Std 802.1AS”;
}
leaf current-compute-neighbor-rate-ratio {
  type boolean;
  config false;
  description
    “Current value of computeNeighborRateRatio.”;
  reference
    “14.10.32 of IEEE Std 802.1AS”;
}
leaf use-mgt-compute-neighbor-rate-ratio {
  type boolean;
```

```
      description
        “Boolean that determines the source of computeNeighborRateRatio..
         If the value is true, computeNeighborRateRatio is set equal to
         the value of mgt-compute-neighbor-rate-ratio. If the value is
         false, computeNeighborRateRatio is determined by the
         LinkDelayIntervalSetting state machine (i.e., changed with
         Signaling message).”;
      reference
        “14.10.33 of IEEE Std 802.1AS”;
    }
    leaf mgt-compute-neighbor-rate-ratio {
      type boolean;
      description
        “Value of computeNeighborRateRatio, used if
         use-mgt-compute-neighbor-rate-ratio is true. This value is not
         used if use-mgt-compute-neighbor-rate-ratio is false.”;
      reference
        “14.10.34 of IEEE Std 802.1AS”;
    }
    leaf initial-compute-mean-link-delay {
      type boolean;
      description
        “When use-mgt-compute-mean-link-delay is false (i.e., change with
         Signaling message), this is the initial value of
         computeMeanLinkDelay.”;
      reference
        “14.10.35 of IEEE Std 802.1AS”;
    }
    leaf current-compute-mean-link-delay {
      type boolean;
      config false;
      description
        “Current value of computeMeanLinkDelay.”;
      reference
        “14.10.36 of IEEE Std 802.1AS”;
    }
    leaf use-mgt-compute-mean-link-delay {
      type boolean;
      description
        “Boolean that determines the source of computeMeanLinkDelay. If
         the value is true, computeMeanLinkDelay is set equal to the value
         of mgt-compute-mean-link-delay. If the value is false,
         computeMeanLinkDelay is determined by the
         LinkDelayIntervalSetting state machine (i.e., changed with
         Signaling message).”;
      reference
        “14.10.37 of IEEE Std 802.1AS”;
    }
    leaf mgt-compute-mean-link-delay {
      type boolean;
      description
        “Value of computeMeanLinkDelay, used if
         use-mgt-compute-mean-link-delay is true. This value is not used
         if use-mgt-compute-mean-link-delay is false.”;
      reference
        “14.10.38 of IEEE Std 802.1AS”;
    }
    leaf allowed-lost-responses {
      type uint8;
```

```
      description
        “Number of Pdelay_Req messages for which a valid response is not
         received, above which a port is considered to not be exchanging
         peer delay messages with its neighbor.”;
      reference
        “14.10.39 of IEEE Std 802.1AS”;
    }
    leaf allowed-faults {
      type uint8;
      description
        “Number of faults above which asCapable is set to false.”;
      reference
        “14.10.40 of IEEE Std 802.1AS”;
    }
    leaf gptp-cap-receipt-timeout {
      type uint8;
      description
        “Number of transmission intervals that a port waits without
         receiving the gPTP capable TLV, before assuming that the neighbor
         port is no longer invoking the gPTP protocol.”;
      reference
        “14.10.41 of IEEE Std 802.1AS”;
    }
    leaf nup {
      type float64;
      description
        “For an OLT port of an IEEE Std 802.3 EPON link, this value is
         the effective index of refraction for the EPON upstream
         wavelength light of the optical path.”;
      reference
        “14.10.43 of IEEE Std 802.1AS”;
    }
    leaf ndown {
      type float64;
      description
        “For an OLT port of an IEEE 802.3 EPON link, this value is the
         effective index of refraction for the EPON downstream wavelength
         light of the optical path.”;
      reference
        “14.10.44 of IEEE Std 802.1AS”;
    }
    leaf one-step-tx-oper {
      type boolean;
      config false;
      description
        “This value is true if the port is sending one-step Sync
         messages, and false if the port is sending two-step Sync and
         Follow-Up messages.”;
      reference
        “14.10.45 of IEEE Std 802.1AS”;
    }
    leaf one-step-receive {
      type boolean;
      config false;
      description
        “This value is true if the port is capable of receiving and
         processing one-step Sync messages.”;
      reference
        “14.10.46 of IEEE Std 802.1AS”;
```

```
    }
    leaf one-step-transmit {
      type boolean;
      config false;
      description
        “This value is true if the port is capable of transmitting
         one-step Sync messages.”;
      reference
        “14.10.47 of IEEE Std 802.1AS”;
    }
    leaf initial-one-step-tx-oper {
      type boolean;
      description
        “When use-mgt-one-step-tx-oper is false (i.e., change with
         Signaling message), this is the initial value of
         current-one-step-tx-oper.”;
      reference
        “14.10.48 of IEEE Std 802.1AS”;
    }
    leaf current-one-step-tx-oper {
      type boolean;
      config false;
      description
        “This value is true if the port is configured to transmit
         one-step Sync messages, either via management
         (mgt-one-step-tx-oper) or Signaling. If both
         current-one-step-tx-oper and one-step-transmit are true, the port
         transmits one-step Sync messages (i.e., one-step-tx-oper true).”;
      reference
        “14.10.49 of IEEE Std 802.1AS”;
    }
    leaf use-mgt-one-step-tx-oper {
      type boolean;
      description
        “Boolean that determines the source of current-one-step-tx-oper.
         If the value is true, current-one-step-tx-oper is set equal to
         the value of mgt-one-step-tx-oper. If the value is false,
         current-one-step-tx-oper is determined by the
         OneStepTxOperSetting state machine (i.e., changed with Signaling
         message).”;
      reference
        “14.10.50 of IEEE Std 802.1AS”;
    }
    leaf mgt-one-step-tx-oper {
      type boolean;
      description
        “If use-mgt-one-step-tx-oper is true, current-one-step-tx-oper is
         set equal to this value. This value is not used if
         use-mgt-one-step-tx-oper is false.”;
      reference
        “14.10.51 of IEEE Std 802.1AS”;
    }
    leaf sync-locked {
      type boolean;
      config false;
      description
        “This value is true if the port will transmit a Sync as soon as
         possible after the timeReceiver port receives a Sync message.”;
      reference
```

```
      "14.10.52 of IEEE Std 802.1AS";
  }
  leaf-list pdelay-truncated-timestamps {
    type uinteger48;
    config false;
    description
      "For full-duplex IEEE Std 802.3 media, and CSN media that use the
       peer-to-peer delay mechanism to measure path delay, the values of
       the four elements of this leaf-list correspond to the timestamps
       t1, t2, t3, and t4, listed in that order. Each timestamp is
       expressed in units of 2^-16 ns (i.e., the value of each array
       element is equal to the remainder obtained upon dividing the
       respective timestamp, expressed in units of 2^-16 ns, by 2^48).
       At any given time, the timestamp values stored in the array are
       for the same, and most recently completed, peer delay message
       exchange. For each timestamp, only 48-bits are valid (the upper
       16-bits are always zero).";
    reference
      "14.10.53 of IEEE Std 802.1AS";
  }
  leaf pdelay-req-send-disabled {
    type boolean;
    description
      "A boolean that is administratively set to TRUE
       if Pdelay_Req messages are not transmitted by this port.
       The default value for this variable shall be FALSE.";
    reference
      "14.10.60 of IEEE Std 802.1AS";
  }
  leaf pdelay-resp-send-disabled {
    type boolean;
    description
      "A boolean that is administratively set to TRUE
       if Pdelay_Resp messages are not transmitted by this port.
       The default value for this variable shall be FALSE.";
    reference
      "14.10.61 of IEEE Std 802.1AS";
  }
}

augment "/ptp-tt:ptp"
      + "/ptp-tt:instances"
      + "/ptp-tt:instance"
      + "/ptp-tt:ports"
      + "/ptp-tt:port" {
  description
    "Augment to add port-statistics-ds to IEEE Std 1588 PTP Port.";
  container port-statistics-ds {
    description
      "Provides counters associated with the port of the PTP Instance.";
    reference
      "14.12 of IEEE Std 802.1AS";
    leaf rx-sync-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time synchronization information
         is received.";
      reference
```

```
          "14.12.2 of IEEE Std 802.1AS";
      }
      leaf rx-one-step-sync-count {
        type yang:counter32;
        config false;
        description
          "Counter that increments every time a one-step Sync message is
           received.";
        reference
          "14.12.3 of IEEE Std 802.1AS";
      }
      leaf rx-follow-up-count {
        type yang:counter32;
        config false;
        description
          "Counter that increments every time a Follow_Up message is
           received.";
        reference
          "14.12.4 of IEEE Std 802.1AS";
      }
      leaf rx-pdelay-req-count {
        type yang:counter32;
        config false;
        description
          "Counter that increments every time a Pdelay_Req message is
           received.";
        reference
          "14.12.5 of IEEE Std 802.1AS";
      }
      leaf rx-pdelay-resp-count {
        type yang:counter32;
        config false;
        description
          "Counter that increments every time a Pdelay_Resp message is
           received.";
        reference
          "14.12.6 of IEEE Std 802.1AS";
      }
      leaf rx-pdelay-resp-follow-up-count {
        type yang:counter32;
        config false;
        description
          "Counter that increments every time a Pdelay_Resp_Follow_Up
           message is received.";
        reference
          "14.12.7 of IEEE Std 802.1AS";
      }
      leaf rx-announce-count {
        type yang:counter32;
        config false;
        description
          "Counter that increments every time an Announce message is
           received.";
        reference
          "14.12.8 of IEEE Std 802.1AS";
      }
      leaf rx-packet-discard-count {
        type yang:counter32;
        config false;
```

```
      description
        “Counter that increments every time a PTP message of the
         respective PTP Instance is discarded.”;
      reference
        “14.12.9 of IEEE Std 802.1AS”;
    }
    leaf sync-receipt-timeout-count {
      type yang:counter32;
      config false;
      description
        “Counter that increments every time a sync receipt timeout
         occurs.”;
      reference
        “14.12.10 of IEEE Std 802.1AS”;
    }
    leaf announce-receipt-timeout-count {
      type yang:counter32;
      config false;
      description
        “Counter that increments every time an announce receipt timeout
         occurs.”;
      reference
        “14.12.11 of IEEE Std 802.1AS”;
    }
    leaf pdelay-allowed-lost-exceeded-count {
      type yang:counter32;
      config false;
      description
        “Counter that increments every time the value of the variable
         lostResponses exceeds the value of the variable
         allowedLostResponses, in the RESET state of the MDPdelayReq
         state machine.”;
      reference
        “14.12.12 of IEEE Std 802.1AS”;
    }
    leaf tx-sync-count {
      type yang:counter32;
      config false;
      description
        “Counter that increments every time synchronization information
         is transmitted.”;
      reference
        “14.12.13 of IEEE Std 802.1AS”;
    }
    leaf tx-one-step-sync-count {
      type yang:counter32;
      config false;
      description
        “Counter that increments every time a one-step Sync message is
         transmitted.”;
      reference
        “14.12.14 of IEEE Std 802.1AS”;
    }
    leaf tx-follow-up-count {
      type yang:counter32;
      config false;
      description
        “Counter that increments every time a Follow_Up message is
         transmitted.”;
```

```
      reference
        "14.12.15 of IEEE Std 802.1AS";
    }
    leaf tx-pdelay-req-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Pdelay_Req message is
         transmitted.";
      reference
        "14.12.16 of IEEE Std 802.1AS";
    }
    leaf tx-pdelay-resp-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Pdelay_Resp message is
         transmitted.";
      reference
        "14.12.17 of IEEE Std 802.1AS";
    }
    leaf tx-pdelay-resp-follow-up-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Pdelay_Resp_Follow_Up
         message is transmitted.";
      reference
        "14.12.18 of IEEE Std 802.1AS";
    }
    leaf tx-announce-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time an Announce message is
         transmitted.";
      reference
        "14.12.19 of IEEE Std 802.1AS";
    }
  }
}

augment "/ptp-tt:ptp"
      + "/ptp-tt:instances"
      + "/ptp-tt:instance"
      + "/ptp-tt:ports"
      + "/ptp-tt:port" {
  description
    "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588 PTP
     Port.";
  container asymmetry-measurement-mode-ds {
    description
      "Represents the capability to enable/disable the Asymmetry
       Compensation Measurement Procedure on a PTP Port. This data set
       is used instead of the CMLDS asymmetry-measurement-mode-ds when
       only a single PTP Instance is present (i.e., CMLDS is not used).";
    reference
      "14.15 of IEEE Std 802.1AS
       Annex G of IEEE Std 802.1AS";
```

```
      leaf enabled {
        type boolean;
        description
          “For full-duplex IEEE Std 802.3 media, the value is true if an
           asymmetry measurement is being performed for the link attached
           to this PTP Port, and false otherwise. For all other media, the
           value shall be false.”;
        reference
          “14.15.2 of IEEE Std 802.1AS”;
      }
    }
  }

  augment “/ptp-tt:ptp”
        + “/ptp-tt:common-services”
        + “/ptp-tt:cmlds”
        + “/ptp-tt:ports”
        + “/ptp-tt:port”
        + “/ptp-tt:link-port-ds” {
    description
      “Augment IEEE Std 1588 cmldsLinkPortDS.”;
    leaf cmlds-link-port-enabled {
      type boolean;
      config false;
      description
        “Boolean that is true if both delay-mechanism is common-p2p and
         the value of ptp-port-enabled is true, for at least one PTP Port
         that uses the CMLDS; otherwise, the value is false.”;
      reference
        “11.2.18.1 of IEEE Std 802.1AS
         14.18.3 of IEEE Std 802.1AS”;
    }
    leaf is-measuring-delay {
      type boolean;
      config false;
      description
        “This leaf is analogous to is-measuring-delay for a PTP Port, but
         applicable to this Link Port.”;
      reference
        “14.18.4 of IEEE Std 802.1AS”;
    }
    leaf as-capable-across-domains {
      type boolean;
      config false;
      description
        “This leaf is true when all PTP Instances (domains) for this Link
         Port detect proper exchange of Pdelay messages.”;
      reference
        “11.2.2 of IEEE Std 802.1AS
         14.18.5 of IEEE Std 802.1AS”;
    }
    leaf mean-link-delay-thresh {
      type ptp-tt:time-interval;
      description
        “Propagation time threshold for mean-link-delay, above which a
         Link Port is not considered capable of participating in the IEEE
         Std 802.1AS protocol.”;
      reference
        “14.18.7 of IEEE Std 802.1AS”;
```

```
    }
    leaf neg-mean-link-delay-thresh {
      type ptp-tt:time-interval;
      description
        "The negative propagation time threshold for mean-link-delay,
         below which a Link Port is not considered capable of
         participating in the IEEE Std 802.1AS protocol.";
      reference
        "14.18.8 of IEEE Std 802.1AS";
    }
    leaf initial-log-pdelay-req-interval {
      type int8;
      description
        "This leaf is analogous to initial-log-pdelay-req-interval for a
         PTP Port, but applicable to this Link Port.";
      reference
        "14.18.11 of IEEE Std 802.1AS";
    }
    leaf current-log-pdelay-req-interval {
      type int8;
      config false;
      description
        "This leaf is analogous to current-log-pdelay-req-interval for a
         PTP Port, but applicable to this Link Port.";
      reference
        "14.18.12 of IEEE Std 802.1AS";
    }
    leaf use-mgt-log-pdelay-req-interval {
      type boolean;
      description
        "This leaf is analogous to use-mgt-log-pdelay-req-interval for a
         PTP Port, but applicable to this Link Port.";
      reference
        "14.18.13 of IEEE Std 802.1AS";
    }
    leaf mgt-log-pdelay-req-interval {
      type int8;
      description
        "This leaf is analogous to mgt-log-pdelay-req-interval for a PTP
         Port, but applicable to this Link Port.";
      reference
        "14.18.14 of IEEE Std 802.1AS";
    }
    leaf initial-compute-neighbor-rate-ratio {
      type boolean;
      description
        "This leaf is analogous to initial-compute-neighbor-rate-ratio
         for a PTP Port, but applicable to this Link Port.";
      reference
        "14.18.15 of IEEE Std 802.1AS";
    }
    leaf current-compute-neighbor-rate-ratio {
      type boolean;
      config false;
      description
        "This leaf is analogous to current-compute-neighbor-rate-ratio
         for a PTP Port, but applicable to this Link Port.";
      reference
        "14.18.16 of IEEE Std 802.1AS";
```

```
    }
    leaf use-mgt-compute-neighbor-rate-ratio {
      type boolean;
      description
        “This leaf is analogous to use-mgt-compute-neighbor-rate-ratio
         for a PTP Port, but applicable to this Link Port.”;
      reference
        “14.18.17 of IEEE Std 802.1AS”;
    }
    leaf mgt-compute-neighbor-rate-ratio {
      type boolean;
      description
        “This leaf is analogous to mgt-compute-neighbor-rate-ratio for a
         PTP Port, but applicable to this Link Port.”;
      reference
        “14.18.18 of IEEE Std 802.1AS”;
    }
    leaf initial-compute-mean-link-delay {
      type boolean;
      description
        “This leaf is analogous to initial-compute-mean-link-delay for a
         PTP Port, but applicable to this Link Port.”;
      reference
        “14.18.19 of IEEE Std 802.1AS”;
    }
    leaf current-compute-mean-link-delay {
      type boolean;
      config false;
      description
        “This leaf is analogous to current-compute-mean-link-delay for a
         PTP Port, but applicable to this Link Port.”;
      reference
        “14.18.20 of IEEE Std 802.1AS”;
    }
    leaf use-mgt-compute-mean-link-delay {
      type boolean;
      description
        “This leaf is analogous to use-mgt-compute-mean-link-delay for a
         PTP Port, but applicable to this Link Port.”;
      reference
        “14.18.21 of IEEE Std 802.1AS”;
    }
    leaf mgt-compute-mean-link-delay {
      type boolean;
      description
        “This leaf is analogous to mgt-compute-mean-link-delay for a PTP
         Port, but applicable to this Link Port.”;
      reference
        “14.18.22 of IEEE Std 802.1AS”;
    }
    leaf allowed-lost-responses {
      type uint8;
      description
        “This leaf is analogous to allowed-lost-responses for a PTP Port,
         but applicable to this Link Port.”;
      reference
        “14.18.23 of IEEE Std 802.1AS”;
    }
    leaf allowed-faults {
```

```
      type uint8;
      description
        “This leaf is analogous to allowed-faults for a PTP Port, but
         applicable to this Link Port.”;
      reference
        “14.18.24 of IEEE Std 802.1AS”;
    }
    leaf-list pdelay-truncated-timestamps {
      type uinteger48;
      config false;
      description
        “This leaf is analogous to pdelay-truncated-timestamps for a PTP
         Port, but applicable to this Link Port.”;
      reference
        “14.18.26 of IEEE Std 802.1AS”;
    }
  }

  augment “/ptp-tt:ptp”
        + “/ptp-tt:common-services”
        + “/ptp-tt:cmlds”
        + “/ptp-tt:ports”
        + “/ptp-tt:port” {
    description
      “Augment to add port-statistics-ds to IEEE Std 1588 Link Port.”;
    container port-statistics-ds {
      description
        “This container is analogous to port-statistics-ds for a PTP
         Port, but applicable to this Link Port.”;
      reference
        “14.19 of IEEE Std 802.1AS”;
      leaf rx-pdelay-req-count {
        type yang:counter32;
        config false;
        description
          “This leaf is analogous to rx-pdelay-req-count for a PTP Port,
           but applicable to this Link Port.”;
        reference
          “14.19.2 of IEEE Std 802.1AS”;
      }
      leaf rx-pdelay-resp-count {
        type yang:counter32;
        config false;
        description
          “This leaf is analogous to rx-pdelay-resp-count for a PTP Port,
           but applicable to this Link Port.”;
        reference
          “14.19.3 of IEEE Std 802.1AS”;
      }
      leaf rx-pdelay-resp-follow-up-count {
        type yang:counter32;
        config false;
        description
          “This leaf is analogous to rx-pdelay-resp-follow-up-count for a
           PTP Port, but applicable to this Link Port.”;
        reference
          “14.19.4 of IEEE Std 802.1AS”;
      }
      leaf rx-packet-discard-count {
```

```
        type yang:counter32;
        config false;
        description
          "This leaf is analogous to rx-packet-discard-count for a PTP
           Port, but applicable to this Link Port.";
        reference
          "14.19.5 of IEEE Std 802.1AS";
      }
      leaf pdelay-allowed-lost-exceeded-count {
        type yang:counter32;
        config false;
        description
          "This leaf is analogous to pdelay-allowed-lost-exceeded-count
           for a PTP Port, but applicable to this Link Port.";
        reference
          "14.19.6 of IEEE Std 802.1AS";
      }
      leaf tx-pdelay-req-count {
        type yang:counter32;
        config false;
        description
          "This leaf is analogous to tx-pdelay-req-count for a PTP Port,
           but applicable to this Link Port.";
        reference
          "14.19.7 of IEEE Std 802.1AS";
      }
      leaf tx-pdelay-resp-count {
        type yang:counter32;
        config false;
        description
          "This leaf is analogous to tx-pdelay-resp-count for a PTP Port,
           but applicable to this Link Port.";
        reference
          "14.19.8 of IEEE Std 802.1AS";
      }
      leaf tx-pdelay-resp-follow-up-count {
        type yang:counter32;
        config false;
        description
          "This leaf is analogous to tx-pdelay-resp-follow-up-count for a
           PTP Port, but applicable to this Link Port.";
        reference
          "14.19.10 of IEEE Std 802.1AS";
      }
    }
  }

  augment "/ptp-tt:ptp"
        + "/ptp-tt:common-services"
        + "/ptp-tt:cmlds"
        + "/ptp-tt:ports"
        + "/ptp-tt:port" {
    description
      "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588 Link
       Port.";
    container asymmetry-measurement-mode-ds {
      description
        "This container is analogous to asymmetry-measurement-mode-ds for
         a PTP Port, but applicable to this Link Port.";
```

```
      reference
        "14.20 of IEEE Std 802.1AS";
      leaf enabled {
        type boolean;
        description
          "This leaf is analogous to
           asymmetry-measurement-mode-ds.enabled for a PTP Port, but
           applicable to this Link Port.";
        reference
          "14.20.2 of IEEE Std 802.1AS";
      }
    }
  }
}
```

Insert the following new Clause 19:

19. Media-dependent layer specification for IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

19.1 Overview

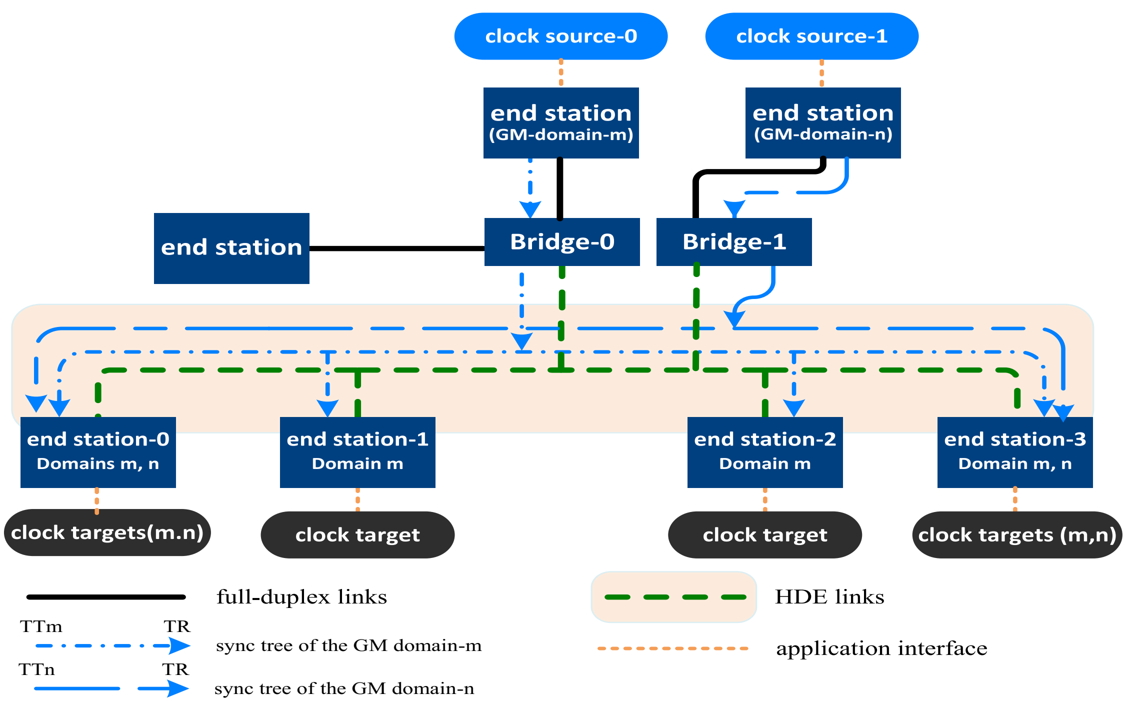
19.1.1 General

This clause specifies the media-dependent layer that provides synchronized time across links using the IEEE Std 802.3 Clause 4 MAC operating in half-duplex mode

19.1.1.1 Half-duplex Ethernet (HDE) characteristics

The IEEE Std 802.3 Clause 4 MAC can operate in either full-duplex or half-duplex mode. When this MAC is operating in full-duplex, its media-dependent specification for gPTP is covered in Clause 11. Clause 19 is used when the IEEE Std 802.3 Clause 4 MAC is operating in half-duplex. This mode necessitates additional managed object settings and frame processing due to the effects of the shared media this mode supports, which are described in this clause.

Figure 19-1 shows an example of time-aware network using HDE links.



Note 1: Bridge-0 and Bridge-1 bridges in this figure are each an example of a time-aware system that contains a PTP Relay Instance, and the end stations with clock targets are examples of time-aware systems that contain PTP End Instances.
Note 2: GM denotes Grandmaster PTP Instance

Figure 19-1— Time-aware network example where the four end stations near the bottom of the figure are connected to the bridge via an HDE link

When using HDE links, the use of "neighbor" refers to a device or to devices (end station or bridge) that is/are the intended recipient(s) of a transmitted PTP message. The intended recipient(s) of a PTP message from a gPTP TT port is/are one or more gPTP TR port(s). The intended recipient of a PTP message from a gPTP TR port is the gPTP TT port, regardless of whether or not other gPTP TT ports received that message (e.g., if all ports are connected via a shared medium). In the example in Figure 19-1, PTP messages are exchanged between a gPTP TT port of a bridge (e.g., Bridge-0) and the four gPTP TR ports of the four end stations via a shared medium.

NOTE—For example, this standard supports 10BASE-T1S, as specified in Clause 147 of IEEE Std 802.3, utilizing Physical Layer Collision Avoidance (PLCA), as specified in Clause 148 of IEEE Std 802.3.

19.1.1.2 Overview of the major differences and restrictions of using HDE links

This standard uses Clause 11 in its entirety with the following differences (complete details follow starting in 19.1.2):

a) The peer delay initiator is restricted to timeReceivers only (see 19.1.2).
b) External port configuration mode is the only mode supported (see 19.8).

The following features of this standard have not been qualified for use with HDE links:

c) One-step time transport (see 19.1.3 and 19.2.16)
d) CMLDS (see 19.2.17)
e) The use of Signaling messages (see 19.8)
f) Hot Standby (see 19.8)

19.1.2 Propagation delay measurement over HDE links

The measurement of propagation delay on an HDE link using the peer-to-peer delay mechanism is illustrated in Figure 11-1 and is described in 11.1.2, with the following differences:

a) The peer delay initiator is restricted to each timeReceiver port
b) The timeTransmitter port does not initiate the peer-to-peer delay mechanism.

Therefore, pdelayReqSendDisabled and pdelayRespSendDisabled are set as follows:

c) pdelayReqSendDisabled is set to FALSE when the PTP Port state is TimeReceiverPort and is otherwise set to TRUE.
d) pdelayRespSendDisabled is set to FALSE when the PTP Port state is TimeTransmitterPort and is otherwise set to TRUE

Other configuration values for item c) and item d) are not intended for use with HDE links by this standard.

19.1.3 Transport of time-synchronization information

HDE links uses two-step time transport as specified in clause 11.1.3. The variables oneStepReceive (see 19.2.13.9), oneStepTransmit (see 19.2.13.10), and oneStepTxOper(see 19.2.13.11) are set to FALSE for HDE links. Other configuration values for oneStepTransmit and oneStepTxOper are not intended for use with HDE links by this standard.

NOTE—The transport of time-synchronization information by a PTP Instance, using Sync and Follow_Up messages, is illustrated in Figure 11-2.

19.1.4 Model of operation

The model for a PTP Instance of a time-aware system with full-duplex point-to-point links is shown in Figure 11-3. This clause reuses Figure 11-3 (as its structure is unchanged for this clause), where all references to clause 11 in Figure 11-3 are to be replaced by references to clause 19 (this clause). The presence of one HDE MD entity per PTP Port is assumed.

A general, media-independent description of the generation of timestamps is given in 8.4.3. A more specific description for PTP event messages is given in 11.3.2.1. A PTP event message is timestamped relative to the LocalClock entity when the message timestamp point (see 3.17) crosses the timestamp measurement plane (see 3.33). The timestamp is corrected for any ingressLatency or egressLatency (see 8.4.3) to produce a timestamp relative to the reference plane (see 3.26). The corrected timestamp value is provided to the MD entity.

The MD entity behavior and detailed state machines specific to full-duplex point-to-point links, which are described in 11.2, are reused for HDE links subject to the conditions defined in 19.2.

19.2 State machines for HDE links

19.2.1 General

The state machines for HDE links are described in 11.2.1 except that there is one instance of the MDPdelayReq state machine per PTP Port, there is one instance of the MDPdelayResp state machine per port, and the LinkDelayIntervalSetting state machine is not used.

19.2.2 Determination of asCapable and asCapableAcrossDomains

As specified in 11.2.2.

NOTE—For full-duplex point-to-point links, asCapableAcrossDomains is one global variable for all domains per link port. For HDE links, asCapableAcrossDomains is a global variable per PTP Instance (i.e., per domain), the name asCapableAcrossDomains has been kept for backwards compatibility with existing implementations.

19.2.3 Use of MAC Control PAUSE operation

As specified in 11.2.3.

19.2.4 Use of priority-based flow control

As specified in 11.2.4.

19.2.5 Use of link aggregation

As specified in 11.2.5.

19.2.6 Service interface primitives and data structures communicated between state machines

The following two subclauses describe the service primitives and data structures communicated between the time-synchronization state machines of the MD entity. First the service primitives are described, followed by the data structures.

19.2.7 DL-UNITDATA.request

This service primitive is as specified in 11.2.7.

19.2.8 DL-UNITDATA.indication

This service primitive is as specified in 11.2.8.

19.2.9 MDTimestampReceive

This data structure is as specified in 11.2.9.

19.2.10 MDSyncReceive

This data structure is as specified in 10.2.2.2.

19.2.11 MDSyncSend

This structure is as specified in 10.2.2.1.

19.2.12 Overview of MD entity global variables

The overview of MD entity global variables is given in 11.2.12.

19.2.13 MD entity global variables

19.2.13.1 currentLogPdelayReqInterval: This variable is as specified in 11.2.13.1. However, for HDE, there is one instance of this variable per PTP Port.

19.2.13.2 initialLogPdelayReqInterval: This variable is as specified in 11.2.13.2. However, for HDE, there is one instance of this variable per PTP Port.

19.2.13.3 pdelayReqInterval: This variable is as specified in 11.2.13.3. However, for HDE, there is one instance of this variable per PTP Port.

19.2.13.4 allowedLostResponses: This variable is as specified in 11.2.13.4. However, for HDE, there is one instance of this variable per PTP Port.

19.2.13.5 allowedFaults: This variable is as specified in 11.2.13.5. However, for HDE, there is one instance of this variable per PTP Port.

19.2.13.6 isMeasuringDelay: This variable is as specified in 11.2.13.6. However, for HDE, the port measures propagation delay if it is receiving Pdelay_Resp and Pdelay_Resp_Follow_Up messages from the requested port with the requested clockIdentity. However, for HDE, there is one instance of this variable per PTP Port.

19.2.13.7 meanLinkDelayThresh: This variable is as specified in 11.2.13.7. However, for HDE, there is one instance of this variable per PTP Port.

19.2.13.8 negMeanLinkDelayThresh: This variable is as specified in 11.2.13.8. However, for HDE, there is one instance of this variable per PTP Port.

19.2.13.9 syncSequenceId: This variable is as specified in 11.2.13.8.

19.2.13.10 oneStepReceive: This variable is as specified in 11.2.13.9. However, for HDE, this variable is FALSE.

19.2.13.11 oneStepTransmit: This variable is as specified in 11.2.13.10. However, for HDE, this variable is FALSE.

19.2.13.12 oneStepTxOper: This variable is as specified in 11.2.13.11.However, for HDE, this variable is FALSE.

19.2.13.13 asCapableAcrossDomains: This variable is as specified in 11.2.13.12.

19.2.13.14 nrrPdelay: This variable is as specified in 11.2.13.13.

19.2.13.15 nrrSync: This variable is as specified in 11.2.13.14.

19.2.13.16 nrrCompMethod: This variable is as specified in 11.2.13.15.

19.2.14 MDSyncReceiveSM state machine

The MDSyncReceiveSM state machine is as specified in 11.2.14.

19.2.15 MDSyncSendSM state machine

The MDSyncSendSM state machine is as specified in 11.2.15.

19.2.16 OneStepTxOperSetting state machine

This state machine is not used for HDE links.

19.2.17 Common Mean Link Delay Service (CMLDS)

CMLDS has not been qualified for use with HDE links.

HDE links use the transport-specific peer-to-peer delay mechanism for each domain. Therefore, if the time-aware system implements multiple domains on an HDE link, each domain uses the transport-specific peer-to-peer delay mechanism on that link regardless of its domain number.

If multiple TimeTransmitter ports are present on an HDE link, they are in different gPTP domains. CMLDS cannot be used because, in general, the TimeTransmitter ports can be on physical ports of different time-aware systems (i.e., different bridges). In this case, both meanLinkDelay and neighborRateRatio between a PTP End Instance in one of the domains and the TimeTransmitter it is communicating with can be different from meanLinkDelay and neighborRateRatio between a PTP End Instance in another domain on the same end station and the TimeTransmitter that PTP End Instance is communicating with. For instance, in Figure 19-1 the meanLinkDelay and neighborRateRatio between the PTP End Instance over domain-m and over domain-n at end station-0 are computed using timestamps from two distinct time aware systems, GM-domain-m and GM-domain-n, respectively; and hence does not use CMLDS.

19.2.18 Common Mean Link Delay Service (CMLDS) global variable

The Common Mean Link Delay Service (CMLDS) global variable is as specified in 11.2.18.

19.2.19 MDPdelayReq state machine

The MDPdelayReq state machine is as specified in 11.2.19.

The variable pdelayReqSendDisabled is set per 19.1.2.

Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmittetd with domainNumber of the HDE gPTP domain used by the transport-specific peer delay mechanism (see 19.4). Any received Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages are ignored by the state machines if the domainNumber in the received message does not match the gPTP domain associated with the PTP Port.

19.2.20 MDPdelayResp state machine

The MDPdelayResp state machine is as specified in 11.2.20.

The variable pdelayRespSendDisabled is set per 19.1.2.

Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmittetd with domainNumber of the HDE gPTP domain used by the transport-specific peer delay mechanism (see 19.4). Any received Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages are ignored by the state machines if the domainNumber in the received message does not match the gPTP domain associated with the PTP Port.

19.2.21 LinkDelayIntervalSetting state machine

This state machine is not used for HDE links.

19.3 Message attributes

Message attributes is as specified in 11.3.

19.4 Message formats

Message formats is as specified in 11.4, except for 11.4.2.4, where the Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmitted with the domainNumber of the HDE gPTP domain used by the transport-specific peer delay mechanism. Any Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up message received with a domainNumber other than the domainNumber of the HDE gPTP domain used by the transport-specific peer delay mechanism shall be ignored. The domainNumber for all other PTP messages is as specified in 10.6.2.2.6.

19.5 Protocol timing characterization

19.5.1 General

This subclause specifies timing attributes for the media-dependent sublayer specified in this clause.

19.5.2 Message transmission intervals

19.5.2.1 General interval specification

The general message transmission interval specification is as specified in 11.5.2.1.

19.5.2.2 Pdelay_Req message transmission interval

Pdelay_Req message transmission interval is as specified in 11.5.2.2. The variable useMgtSettableLogPdelayReqInterval is set to TRUE. Other configuration values for useMgtSettableLogPdelayReqInterval are not intended for use with HDE links by this standard.

19.5.2.3 Sync message transmission interval default value

The Sync message transmission interval default value is as specified in 11.5.2.3.

19.5.3 allowedLostResponses

The variable allowedLostResponses is as specified in 11.5.3.

19.5.4 allowedFaults

The variable allowedFaults is as specified in 11.5.4.

19.6 Control of computation of neighborRateRatio

The control of computation of neighborRateRatio is as specified in 11.6.

19.7 Control of computation of meanLinkDelay

The control of computation of meanLinkDelay is as specified in 11.7.

19.8 HDE links settings and configuration

This subclause provides settings and configurations that are specific for HDE links, other configuration values are not intended for use with HDE links by this standard.

a) The per PTP Instance global variable externalPortConfigurationEnabled is set to TRUE for HDE links.
b) Both GptpCapableTransmit and GptpCapableReceive state machines are disabled for HDE links, and therefore gptpCapableStateMachinesEnabled are set to FALSE (see 10.4.1). According to 10.4.1, if the managed object gptpCapableStateMachinesEnabled is FALSE, the global variable neighborGptpCapable for the port (see 10.2.5.16) is set to TRUE.
c) The variables useMgtSettableLogSyncInterval and useMgtSettableLogAnnounceInterval are set to TRUE. As a consequence, the SyncIntervalSetting and the AnnounceIntervalSetting state machines are not used for HDE links.
d) The GptpCapableIntervalSetting state machine is not used for HDE links. The variable useMgtSettableLogGptpCapableMessageInterval is set to TRUE.

Hot Standby (see Clause 18) has not been qualified for use with HDE links.

Annex A

(normative)

Protocol Implementation Conformance Statement (PICS) proforma[3]

Add a row at the end of Table A.5 as follows:

A.5 Major Capabilities

Item	Feature	Status	References	Support
MDHDE	Does the PTP Instance support media-dependent HDE link functionality on one or more PTP Ports?	O.1	5.9, 11, 19, A.6, A.22	Yes [ ] No [ ]

Change A.6 as follows:

A.6 Media access control methods

Item	Feature	Status	References	Support
MAC-IEEE-802.3 MAC-IEEE-802.11	Which MAC methods are implemented in conformance with the relevant MAC standards?	O:2 O:2	11.1 19.1 12.1	Yes [ ] No [ ] Yes [ ] No [ ] Yes [ ] No [ ]
MAC-1	Has a PICS been completed for each of the MAC methods implemented as required by the relevant MAC Standards?	M		Yes [ ]
MAC-2	Do all the MAC methods implemented support the MAC Timing aware Service as specified?	M	Clause 11 Clause 12 Clause 13 Clause 19	Yes [ ]

[3] *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

Change A.21 as follows:

A.21 External port configuration

Item	Feature	Status	References	Support
	If item EXT is not supported, mark N/A.			NA[ ]
EXT-1	Does the PTP Instance support the specifications for externalPortConfigurationEnabled value of true?	EXT:M	10.3.1 19.8	Yes [ ]
EXT-2	Does the PTP Instance support the PortAnnounceInformationExt state machine?	EXT:M	10.3.14	Yes [ ]
EXT-3	Does the PTP Instance support the PortStateSettingExt state machine?	EXT:M	10.3.15	Yes [ ]

<<Editor’s note: Table A.22 is based on Table A.13 for full-duplex point-to-point link. Please, check.>

Insert Table A.22 as follows:

A.22 Media-dependent, HDE link

Item	Feature	Status	References	Support
MDHDE-1	Does this PTP Port implement the functionality of the MDSyncReceiveSM state machine in compliance with the requirements of 19.2.14, 11.2.14 and Figure 11-6?	MDHDE:M	19.2.14, 11.2.14	Yes [ ]
MDHDE-2	Does this PTP Port implement the functionality of the MDSyncSendSM state machine in compliance with the requirements of 19.2.15, 11.2.15 and Figure 11-7?	MIMSTR and MDHDE:M	19.2.15, 11.2.15	Yes [ ]
MDHDE-3	Does this port implement the functionality of the MDPdelayRequest state machine in compliance with the requirements of 19.1.2, 19.2.19, 11.2.19 and Figure 11-9?	MDHDE:M	19.1.2, 19.2.19, 11.2.19	Yes [ ]
MDHDE-4	Does this port implement the functionality of the MDPdelayResponse state machine in compliance with the requirements of 19.1.2, 19.2.20, 11.2.20 and Figure 11-10?	MDHDE:M,	19.2.20 19.1.2 11.2.20	Yes [ ]
MDHDE-5	Does this PTP Port timestamp Sync messages on ingress with respect to the LocalClock in compliance with 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes [ ]
MDHDE-6	Does this PTP Port timestamp Sync messages on egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MIMSTR and MDHDE:M	19.3 11.3.2.1	Yes [ ]
MDHDE-7	Does this port timestamp Pdelay_Req messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes [ ]

A.22 Media-dependent, HDE link *(continued)*

Item	Feature	Status	References	Support
MDHDE-8	Does this port timestamp Pdelay_Resp messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes [ ]
MDHDE-9	Are all IEEE 802.1AS messages on this port sent without a Q-tag in compliance with the requirements of 19.3, 11.3.3?	MDHDE:M	19.3, 11.3.3	Yes [ ]
MDHDE-10	Do all media-dependent messages transmitted on this port use a destination MAC address taken from Table 11-3 in compliance with the requirements of 19.3 and 11.3.4 [01-80-C2-00-00-0E]?	MDHDE:M	19.3, 11.3.4	Yes [ ]
MDHDE-11	Do all media-dependent messages transmitted on this port use a source MAC address that is assigned to that port in compliance with the requirements of 19.3 and 11.3.4?	MDHDE:M	19.3, 11.3.4	Yes [ ]
MDHDE-12	Do all media-dependent message tranmitted on this port us an EtherType specified in Table 11-4 [88-F7]?	MDHDE:M	19.3, 11.3.5	Yes [ ]
MDHDE-13	Does the header of all the media-dependent messages on this port comply with the requirements of 19.4, 11.4.2 and Table 10-7?	MDHDE:M	19.4, 11.4.2	Yes [ ] N/A [ ]
MDHDE-14	Does the body of Sync messages sent on this PTP Port comply with the requirements of 19.4, 11.4.3, Table 11-8, and Table 11-9?	MDHDE:M	19.4, 11.4.3	Yes [ ]
MDHDE-15	Does the body of Follow_Up messages sent on this PTP Port comply with the requirements of 19.4, 11.4.4, 6.4.3.3 (lastGmPhaseChange), and Table 11-10?	MDHDE:M	19.4, 11.4.4, 6.4.3.3	Yes [ ]
MDHDE-16	Does the body of Pdelay_Req messages sent on this port comply with the requirements of 19.4, 11.4.5 and Table 11-12?	MDHDE:M	19.4, 11.4.5	Yes [ ]
MDHDE-17	Does the body of Pdelay_Resp messages sent on this port comply with the requirements of 19.4, 11.4.6 and Table 11-13?	MDHDE:M	19.4, 11.4.6	Yes [ ]
MDHDE-18	Does the body of Pdelay_Resp_Follow_Up messages sent on this port comply with the requirements of 19.4, 11.4.7 and Table 11-14?	MDHDE:M	19.4, 11.4.7	Yes [ ]
MDHDE-19	Are all reserved fields in media-dependent messages sent on this port set to 0 in compliance with the requirements of 19.4, 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes [ ]
MDHDE-20	Do the Sync message sequence numbers comply with the requirements of 19.3, 11.3.8?	MIMSTR and MDHDE:M	19.3, 11.3.8	Yes [ ] N/A [ ]

A.22 Media-dependent, HDE link *(continued)*

Item	Feature	Status	References	Support
MDHDE-21	Do the Pdelay_Req message sequence numbers comply with the requirements of 19.3 and 11.3.8?	MDHDE:M	19.3, 11.3.8	Yes [ ]
MDHDE-22	Does the Pdelay mean request transmission interval comply with the requirements of 19.5.2.2 and 11.5.2.2?	MDHDE:M	19.5.2.2, 11.5.2.2	Yes [ ]
MDHDE-23	Does the Sync mean transmission interval comply with the requirements of 19.5.2.3 and 11.5.2.3?	MDHDE:M	19.5.2.3, 11.5.2.3	Yes [ ]
MDHDE-24	Does the HDE media-dependent layer set the asCapable global variable in the media-independent PortSync entity in compliance with the requirements of 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes [ ]
MDHDE-25	Does the PTP Instance consider the PTP Port or Link Port, respectively, to not be exchanging Pdelay messages when a valid response is not received in compliance with the requirements of 19.5.3 and 11.5.3?	MDHDE:M	19.5.3, 11.5.3	Yes [ ]
MDHDE-26	Does the PTP Instance ignore TLVs, of PTP messages, that it cannot parse and attempt to parse the next TLV, in compliance with the requirements of 19.4 and 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes [ ]
MDHDE-27	Does the time-aware system initialize meanLinkDelayThresh as specified in 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes [ ]
MDHDE-28	Does this port support propagation delay averaging?	MDHDE:O	19.2.19, 11.2.19.3.4	Yes [ ] No [ ]
MDHDE-29	Does this port support two-step capability on receive?	MDHDE:M	19.2.14, 11.2.14, item d) of 5.5	Yes [ ]
MDHDE-30	Does this port support two-step capability on transmit?	MDHDE:M	19.2.15, 11.2.15, item e) of 5.5	Yes [ ]
MDHDE-31	Is the transport-specific peer-to-peer delay mechanism supported?	MDHDE:M	19.2.17	Yes [ ]

Annex F

(informative)

PTP profile included in this standard

F.1 General

Change the paragraph in F.1 as follows:

The specification in this standard of synchronized time transport over a full-duplex point-to-point link includes a PTP profile. The information contained in a PTP profile is described in 20.3 of IEEE Std 1588-2019. This annex summarizes the PTP profile for transport of timing over full-duplex point-to-point links. This PTP profile is also used in the transport of timing over HDE links (see Clause 19), and in the transport of timing over CSN when a CSN clock reference is not present (see Clause 16). This PTP profile is not used in the transport of timing over IEEE Std 802.11 links and IEEE Std 802.3 EPON links; both these transports use native timing mechanisms to assist in the synchronized time transport.